

Review

LLMs in Cyber Security: Bridging Practice and Education

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Abstract

Large Language Models (LLMs) have emerged as powerful tools in cyber security, enabling automation, threat detection, and adaptive learning. Their ability to process unstructured data and generate context-aware outputs supports both operational tasks and educational initiatives. Despite their growing adoption, current research often focuses on isolated applications, lacking a systematic understanding of how LLMs align with domain-specific requirements and pedagogical effectiveness. This highlights a pressing need for comprehensive evaluations that address the challenges of integration, generalization, and ethical deployment in both operational and educational cyber security environments. Therefore, this paper provides a comprehensive and State-of-the-Art review of the significant role of LLMs in cyber security, addressing both operational and educational dimensions. It introduces a holistic framework that categorizes LLM applications into six key cyber security domains, examining each in depth to demonstrate their impact on automation, context-aware reasoning, and adaptability to emerging threats. The paper highlights the potential of LLMs to enhance operational performance and educational effectiveness while also exploring emerging technical, ethical, and security challenges. The paper also uniquely addresses the underexamined area of LLMs in cyber security education by reviewing recent studies and illustrating how these models support personalized learning, hands-on training, and awareness initiatives. The key findings reveal that while LLMs offer significant potential in automating tasks and enabling personalized learning, challenges remain in model generalization, ethical deployment, and production readiness. Finally, the paper discusses open issues and future research directions for the application of LLMs in both operational and educational contexts. This paper serves as a valuable reference for researchers, educators, and practitioners aiming to develop intelligent, adaptive, scalable, and ethically responsible LLM-based cyber security solutions.



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Keywords: Large Language Models (LLMs); cyber security; cyber security education; generative AI; privacy; ethics; AI in cyber security

1. Introduction

Large Language Models (LLMs) have emerged as powerful tools across a wide range of fields, revolutionizing the way information is processed, generated, and understood. In domains such as healthcare, finance, education, cyber security, and software engineering, LLMs enhance productivity by automating complex tasks, generating human-like text, and enabling advanced data analysis [1]. Their ability to understand context, synthesize large volumes of information, and produce coherent responses has led to advances in personalized learning, intelligent tutoring, automated report writing, code generation, and even clinical decision support [2]. LLMs also foster innovation by bridging knowledge gaps and democratizing access to expertise. Their scalability and adaptability make them

invaluable for both individual users and large organizations, positioning them as key enablers of efficiency, accessibility, and creative problem-solving in the digital age [2,3].

The cyber security domain faces persistent challenges, such as an evolving threat landscape, a global skills shortage, and the increasing sophistication of cyberattacks. Traditional tools often struggle to keep pace with zero-day exploits, social engineering tactics, and large-scale data breaches [4]. In response to these issues, LLMs have emerged as powerful tools, offering significant impact and benefits across the cyber security lifecycle. LLMs can accelerate threat intelligence by extracting insights from unstructured data, generating indicators of compromise (IOCs), and summarizing security reports in real-time [5]. They assist in vulnerability assessment, automate incident response workflows, and support hands-on training through interactive simulations and personalized guidance. Furthermore, LLMs enhance security awareness initiatives by delivering tailored educational content to diverse user profiles.

Recent statistics highlight the rapidly growing integration of LLMs into cyber security and education, where more than 55% of organizations have adopted generative AI tools, including LLMs, for cyber security operations, such as threat detection and incident response, and 82% plan to increase their investment in AI-driven security solutions over the next two years [6]. The global market for LLMs in cyber security is projected to exceed USD 11.2 billion by 2033, driven by increasing demand for scalable, intelligent defence mechanisms [6]. In parallel, LLM adoption in education is also accelerating, where AI tools are being used to support personalized learning, simulate attacks in safe environments, and train students in ethical hacking and incident responses [7].

The integration of LLMs into cyber security presents a complex and evolving research problem, shaped by dynamic threat landscapes, data sensitivity, and ethical concerns. While recent studies have examined LLM applications in areas such as threat intelligence, malware analysis, penetration testing, and incident response, most focus on isolated tasks and overlook how LLMs align with domain-specific requirements, data constraints, or operational workflows [7,8]. Similarly, recent work on LLMs in cyber security education often lacks depth, with limited attention to instructional effectiveness, long-term learning outcomes, or ethical use in the classroom setting [9]. To date, existing reviews have largely overlooked this dual perspective, offering either narrowly scoped technical surveys or education-focused discussions that do not reflect operational complexities. This fragmented landscape reveals a clear gap, which is the absence of a comprehensive and integrative review that critically examines LLM applications across both operational and educational domains while addressing technical, pedagogical, and ethical dimensions. As interest in LLMs grows, there is a critical need to move beyond siloed research and examine their role across both operational and educational contexts in an integrated manner. Our study is motivated by this need for a comprehensive understanding that bridges technological innovation with educational and ethical imperatives.

This paper provides a comprehensive and State-of-the-Art review of the evolving role of LLMs in cyber security, addressing both operational and educational dimensions. It goes beyond the existing literature by offering a holistic framework that categorizes LLM applications into six major cyber security domains, with each domain examined in depth to highlight the significance of LLMs in context-aware reasoning, automation, and adaptability to emerging threats. The paper also presents various opportunities for applying LLMs across key cyber security domains, demonstrating their potential to enhance both operational performance and educational effectiveness. Moreover, it identifies and discusses emerging challenges related to LLM integration in cyber environments. The paper uniquely addresses the underexamined area of LLMs in cyber security education by reviewing recent studies and highlighting how these models support personalized

learning, hands-on training, and awareness initiatives. Finally, the paper discusses open issues and future research directions for the application of LLMs in both operational and educational contexts.

Compared to similar reviews, this paper offers a uniquely comprehensive perspective on the application of LLMs in cyber security by bridging both operational and educational dimensions, which is an approach not previously explored in the existing literature. While prior surveys have focused primarily on the technical implementation of LLMs in specific security domains, this paper stands out by introducing a holistic framework that systematically categorizes LLM applications across six critical areas of cyber security. This framework not only clarifies the diverse roles LLMs play in automating tasks, enabling context-aware reasoning, and responding to emerging threats but also provides a structured lens through which to assess their broader impact. In addition, this is the first review to explore the significant potential of LLMs in cyber security education, examining how these models enhance personalized learning, hands-on training, and awareness initiatives. By integrating both practical deployments and pedagogical strategies, the paper addresses a critical gap in current research and sets a foundation for future interdisciplinary research.

The contributions of this paper can be summarized as follows:

1. Provides a comprehensive, State-of-the-Art review of LLM applications in cyber security, spanning both operational and educational domains.
2. Proposes a holistic framework that systematically categorizes LLM use across six core cyber security domains, highlighting their roles in automation, context-aware reasoning, and adaptability to evolving threats.
3. Explores diverse practical opportunities for leveraging LLMs across various cyber security functions, highlighting their emerging capabilities.
4. Identifies and critically examines emerging challenges in integrating LLMs into cyber environments, including technical, ethical, and security-related implications.
5. Provides an in-depth review of LLMs in cyber security education, analysing their applications in personalized learning, interactive training, and awareness-building while discussing key challenges.
6. Outlines open issues and suggests future research directions for the responsible and effective use of LLMs in both cyber security practice and education.

The rest of this paper is structured as follows: Section 2 presents an overview of LLMs, their architecture, and the evolution of AI/ML in cyber security; Section 3 compares traditional cyber security systems with those incorporating LLMs; Section 4 reviews recent studies applying LLMs across various cyber security domains, while Section 5 explores their opportunities and practical applications. Section 6 outlines key challenges in adopting LLMs for cyber security. Section 7 focuses on LLMs in cyber security education, highlighting their benefits, applications, and challenges. Section 8 discusses open issues and future research directions for the use of LLMs in cyber security for both operational and educational dimensions, and Section 9 concludes the paper.

2. Background

This section provides an overview of LLMs, their technical foundations, and how advancements in AI and ML have historically shaped cyber security practices.

2.1. An Overview of LLMs

LLMs have greatly impacted Natural Language Processing (NLP), enabling machines to comprehend and generate human-like text with remarkable fluency. These models are trained on vast amounts of data encompassing diverse linguistic patterns and knowledge domains [10]. Prominent examples such as GPT-4, PaLM, and LLaMA have demonstrated

proficiency across a spectrum of tasks, including text summarization, translation, and code generation [11]. The evolution of LLMs has been driven by advancements in computational resources and the availability of extensive training data, adhering to scaling laws that correlate model performance with size and data volume [12,13].

The architecture of LLMs is primarily based on the transformer model, introduced by Vaswani et al. [14], which relies on self-attention mechanisms to process sequential data efficiently. Unlike traditional RNNs (Recurrent Neural Networks) or LSTMs (Long Short-Term Memory), transformers process entire sequences in parallel, making them highly scalable for large datasets. The architecture consists of an encoder–decoder structure with multiple layers of attention and feedforward neural networks [15]. The essential components of LLM architecture include multi-head attention, positional encoding, and layer normalization. Multi-head attention allows the model to focus on different parts of the input simultaneously, capturing complex linguistic patterns. Positional encoding ensures the model retains information about word order, while layer normalization stabilizes training. Modern LLMs, like GPT-4 and PaLM, scale these components to billions of parameters, enhancing their ability to generate coherent and contextually relevant text [16,17].

Several researchers have proposed various architectures and layer configurations for LLMs [14,18,19]. However, the most common architecture consists of six layers, as shown in Figure 1, as follows:

I. Input Layer: Tokenization and Embeddings

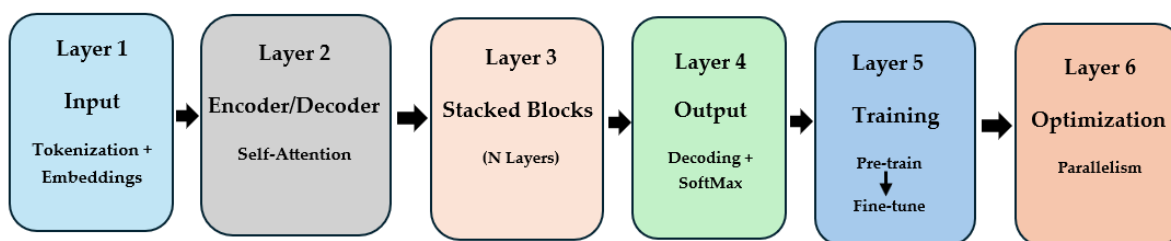


Figure 1. Layered architecture of LLMs.

LLMs begin by transforming raw text into machine-readable tokens, words, or sub-words through tokenization. These tokens are converted into semantic vector representations using word embeddings. To capture word order, positional embeddings are added, compensating for the transformer’s lack of inherent sequential awareness [20].

II. Encoder/Decoder Layer

At the core of LLMs are transformer encoder and decoder layers that utilize self-attention to capture contextual relationships among tokens. Using queries, keys, and values, the model identifies relevant words and computes attention weights. Multi-head attention allows parallel context capture, while feedforward networks, residual connections, and layer normalization enhance stability and performance [18].

III. Stacking of Layers (Transformer Blocks)

Transformer layers are stacked to form deep networks, enabling hierarchical language understanding. As depth increases, the model captures increasingly abstract and complex patterns, facilitating tasks from grammar recognition to advanced reasoning and dialogue handling [14].

IV. Output Layer: Decoding and Prediction

The output layer generates predictions from internal representations. Autoregressive models, like GPT, predict the next token in a sequence, while masked models, like BERT,

fill in missing tokens. A softmax function generates probability distributions over the vocabulary, allowing outputs for tasks like completion and question-answering [10].

V. Training and Fine-Tuning Layers

LLMs are pretrained on large text corpora to learn general language patterns using objectives like masked or next-token prediction. They are then fine-tuned on specific tasks, adjusting model weights and hyperparameters to enhance performance in targeted applications, such as summarization or translation [11,12].

VI. Optimization and Scaling Layers

Given their scale, LLMs require significant computational resources. Gradient descent minimizes loss during training, while techniques like model/data parallelism support scalability. Inference efficiency is improved through methods such as quantization, pruning, and distillation, reducing resource demands without sacrificing accuracy [12].

2.2. History of AI/ML in Cyber Security

The integration of Artificial Intelligence (AI) and Machine Learning (ML) into cyber security has evolved significantly over the past two decades, driven by the increasing scale, complexity, and sophistication of cyber threats. Traditional cyber security mechanisms, primarily rule-based systems, were effective in earlier computing environments where threats were relatively predictable and less dynamic. However, the surge in zero-day vulnerabilities, polymorphic malware, and targeted attacks necessitated more adaptive and intelligent defence mechanisms, giving rise to the adoption of AI/ML techniques [4]. Initial applications of ML in cyber security emerged in the early 2000s, particularly in spam filtering and Intrusion Detection Systems (IDSs). One of the earliest successful applications was the use of Naïve Bayes (NB) and Support Vector Machines (SVM) for spam classification, as demonstrated by Sahami et al. [21]. These early models laid the groundwork for anomaly detection techniques, where ML models learned normal network behaviour and flagged deviations as potential threats [22]. As cyber threats evolved, the need for more complex models became a necessity. Researchers began applying unsupervised learning and clustering techniques for malware detection and behavioural analysis [23].

The 2010s witnessed a shift towards Deep Learning (DL) techniques, enabled by advances in computational power and large-scale datasets. RNNs and Convolutional Neural Networks (CNNs) found applications in detecting botnets, network intrusions, and malware families [24,25]. These models offered the advantage of learning hierarchical representations from raw data, significantly improving detection accuracy. For example, Saxe and Berlin [25] introduced a DL network for static malware classification that outperformed traditional feature engineering approaches. Despite these advances, AI/ML in cyber security also presented challenges. Adversarial ML became an area of growing concern, highlighting the vulnerability of ML models to evasion and poisoning attacks [26]. Additionally, the explainability of AI systems remained a key issue, particularly in critical domains like threat detection and incident response. In recent years, the emergence of transfer learning and pretrained models has opened new frontiers in cyber security applications, with models like BERT and GPT being adapted for threat intelligence, phishing detection, and code analysis [27]. These developments laid the foundation for today's exploration of LLMs in cyber security, enabling a more contextual, human-like understanding of threats, behaviours, and mitigations.

While each generation of AI/ML approaches in cyber security has contributed notable advancements, their evolution also reflects a series of trade-offs. Traditional ML models, such as SVMs and decision trees, offered transparency and efficiency but required extensive feature engineering and struggled with adaptability to novel threats. DL techniques

improved detection accuracy and scalability by learning hierarchical features; however, they lacked interpretability and introduced high computational demands [26]. More recently, pretrained language models and LLMs have demonstrated strong performance in handling unstructured data and performing complex reasoning tasks. However, their use introduces new concerns, including susceptibility to adversarial prompts, hallucination, and ethical risks related to bias and misuse [27]. Thus, while the effectiveness of AI approaches has generally increased over time, so too have their limitations, requiring careful consideration of robustness, transparency, and governance in modern deployments. This trajectory highlights the need for critical evaluation as AI systems are integrated into increasingly sensitive cyber environments (Figure 2).

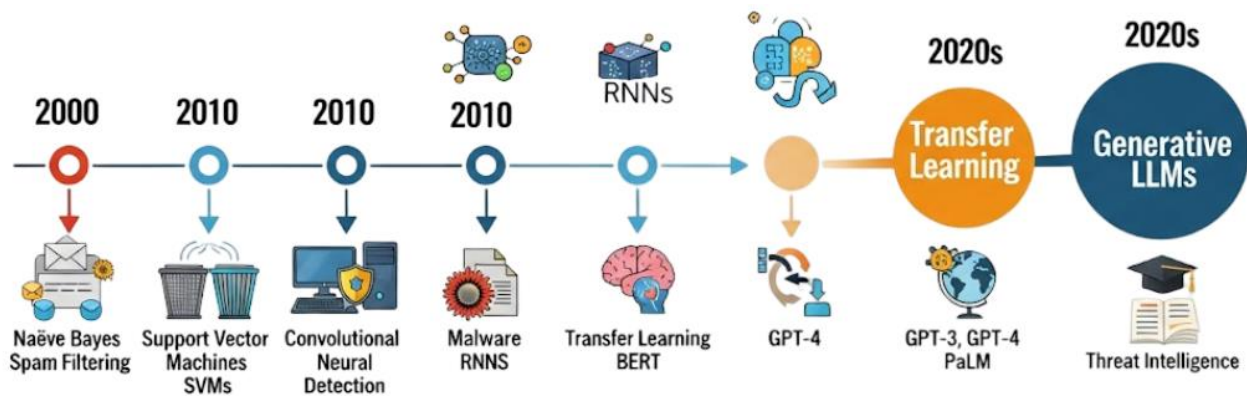


Figure 2. Evolution of AI and ML in cyber security leading to LLMs [13].

3. Traditional vs. LLM-Based Cyber Security

Traditional cyber security systems have long relied on deterministic, rule-based mechanisms and statistical ML approaches to detect and mitigate cyber threats. These methods have been instrumental in protecting against a wide range of attacks, such as malware, phishing, network intrusions, and Denial-of-Service (DoS) attacks. Signature-based systems, such as antivirus software and IDSs, are built on predefined patterns of known threats. While effective against well-understood and previously identified attacks, these systems often struggle to adapt to novel or obfuscated threats [28]. Beyond signature-based systems, traditional anomaly detection techniques leverage statistical methods or classical ML algorithms like Decision Trees (DTs), SVMs, and K-nearest neighbours (KNNs). These systems require extensive feature engineering and domain knowledge to design inputs that represent security events, such as packet payloads or user behaviour patterns [29]. While these approaches can identify deviations from normal behaviour, they are often limited by false positives, lack of scalability, and reduced adaptability in dynamic and adversarial environments. Although these approaches have demonstrated effectiveness in controlled environments, empirical studies have increasingly shown their limitations in dynamic or adversarial settings.

Moreover, traditional cyber security tools typically operate in siloed architectures, focusing on narrow domains, such as endpoint protection, firewall filtering, or network traffic analysis. They lack a holistic understanding across contexts, languages, and modalities, making it difficult to detect complex multi-stage attacks or interpret intent in unstructured data like emails, chats, or social media posts [30]. LLMs represent a paradigm shift in cyber security applications. Unlike traditional models that rely on handcrafted features and limited context, LLMs are pretrained on massive amounts of data and fine-tuned to understand complex linguistic and semantic structures. This enables LLMs to excel in natural language understanding, making them particularly effective in tasks such as phishing

detection, threat intelligence summarization, vulnerability triage, and social engineering simulation [31].

A key differentiator of LLMs is their contextual reasoning ability. They can generate, interpret, and summarize text-based artefacts such as logs, code snippets, or threat reports, which traditional systems cannot handle natively. For example, LLMs can analyse an email and determine whether its tone, structure, and wording match common phishing tactics, without the need for explicit rules or labelled datasets [32]. Similarly, in malware analysis, LLMs can assist in deobfuscating code and suggesting interpretations or remediations, functioning as intelligent assistants rather than just alert generators. Another major advantage of LLMs is their adaptability through few-shot and zero-shot learning, where the model performs tasks without extensive retraining. This is crucial in cyber security, where new threats emerge constantly, and labelled data is rare [33].

Additionally, LLMs facilitate the automation of routine but time-consuming tasks that typically require human intervention, such as generating policy reports, summarizing lengthy security logs, and even writing secure code snippets. Traditional systems, while capable of identifying known threat patterns, often require significant analyst involvement to interpret and act on results. LLMs can streamline this process by offering actionable insights or recommended mitigations in natural language, significantly reducing response time in Security Operations Centres (SOCs) [34]. Furthermore, their ability to interact conversationally allows for seamless integration with analysts through chatbots or co-pilot tools, enabling real-time assistance and bridging the gap between data complexity and human interpretation [35]. Table 1 compares traditional cyber security systems and LLM-based systems.

Table 1. Comparison of traditional cyber security systems vs. LLM-based systems.

Criteria	Traditional Cyber Security Systems	LLM-Based Systems
Approach	Rule-based, statistical, feature-engineered [28].	Context-aware, semantic, and generative [32].
Data Requirements	Labelled datasets and extensive feature engineering [30].	Can operate with unlabelled data and support few/zero-shot learning [32].
Scope	Specific domains.	Multi-domains [1].
Handling NLP	Limited NLP capabilities [29].	Native understanding and generation of text [10].
Adaptability to New Threats	Slow. Requires manual updates or retraining [28].	Fast. Can generalize across tasks and adapt rapidly [7].
Explainability	Limited. Often requires external interpretation [28].	Moderate; outputs are interpretable but not always explainable [33].
Adversarial Exploits	Low but vulnerable to obfuscation.	High due to prompt injection, hallucination, and dual-use concerns [7].
Integration with Analysts	Provides alerts or logs.	Can serve as intelligent assistants or co-pilots.
Scalability	Limited by predefined rules and infrastructure [29].	Highly scalable due to cloud-based inference and parallel processing.
Real-Time Analysis	Effective for known signatures but struggles with novel attacks [28].	Can process and correlate unstructured data in near-real time [34].

Table 1. Cont.

Criteria	Traditional Cyber Security Systems	LLM-Based Systems
False Positive/Negative Rate	High false positives due to rigid rules [30].	Context-aware filtering can reduce false positives but may introduce hallucinations [32].
Cost of Deployment	Moderate (hardware/software licenses).	Potentially high (API costs and GPU compute) but flexible pricing models [35].
Maintenance Overheads	Requires frequent updates to rules/signatures [29].	Self-improving via fine-tuning but needs oversight for drift/hallucinations [34].
Ethical Risks	Well-understood regulatory alignment.	Emerging concerns, like bias, privacy, or misuse of generated content [32].
Examples	Snort IDS [28], ClamAV [30], NetFlow analysers [30].	GPT-4 for phishing detection [33], Copilot for secure coding [33], LLaMA-Guard [36], among others.

While Table 1 offers a conceptual comparison, it is also important to anchor these contrasts in real-world evaluations. Several empirical studies provide insights into the comparative performance of traditional and LLM-based approaches. For instance, Noever [37] demonstrated that GPT-4 detected approximately four times more vulnerabilities than traditional static analysers, such as Fortify and Snyk, while also proposing viable remediations. In a separate study, Sheng et al. [7] showed that LLM-driven systems, like LProtector, outperformed baseline vulnerability detection tools on C/C++ codebases in terms of F1 score, demonstrating higher accuracy and recall. Similarly, Nahmias et al. [38] reported that LLMs trained on contextual persuasion features outperformed classical classifiers in spear-phishing detection, suggesting strong capability in identifying subtle semantic cues that traditional filters miss.

Although LLMs provide countless benefits, the integration of LLMs into cyber security workflows must be approached with caution. While they augment human capabilities, they also introduce new dependencies and vulnerabilities. For instance, overreliance on model outputs without verification may lead to overlooked threats or the implementation of flawed mitigations. Moreover, the ethical concerns around bias, data leakage, and misuse of generative outputs necessitate strong governance frameworks [36]. These concerns are even more noticeable in critical infrastructure environments, where real-time processing constraints, regulatory compliance requirements, and the need for transparent, explainable decision-making limit the viability of opaque, high-latency models. As a result, the deployment of LLMs in such contexts demands additional scrutiny, human oversight, and integration with trusted, auditable systems. In addition, as highlighted by Sovrano et al. [39], LLMs exhibit position-based bias, frequently missing vulnerabilities that appear near the end of input sequences, a flaw not typically observed in deterministic systems. Moreover, empirical evaluations by Isozaki et al. [40] and Happe et al. [41] found that while LLMs like GPT-4-turbo could automate penetration testing sub-tasks, they still struggled with multi-step reasoning and error recovery, especially without human intervention. These findings highlight that while LLMs offer promising new capabilities, they are not universally superior and often require complementary integration with traditional methods. Therefore, a hybrid approach that combines the precision and predictability of traditional systems with the flexibility and intelligence of LLMs is currently the most pragmatic pathway forward.

Building on this hybrid approach, a growing number of industry solutions are adopting hybrid models that combine both paradigms. Traditional tools, such as signature-based

IDSs or rule-based firewalls [28], provide reliability, interpretability, and regulatory familiarity. However, they lack adaptability to novel, unstructured, or evolving threats [30]. On the other hand, LLMs offer enhanced context awareness and generative capabilities, excelling in natural language understanding and real-time threat analysis [32,34]. Hybrid architectures aim to utilize the strengths of both: using deterministic models for baseline threat detection while leveraging LLMs for tasks like log summarization, phishing analysis, or contextual alert prioritization [33,35]. This layered approach not only improves detection accuracy but also addresses critical operational concerns, such as explainability and analyst trust. Industry implementations, including LLM-assisted co-pilots integrated into SOCs [34], reflect this trend and point toward a pragmatic path forward where legacy systems and AI-powered tools merge in a secure, scalable manner.

4. LLMs in Cyber Security

The increasing adoption of LLMs in cyber security reflects a notable shift in how modern threats are detected, understood, and mitigated. As cyber threats grow in scale, complexity, and subtlety, traditional systems often fall short in responding to evolving attack vectors, particularly those that leverage social engineering, code obfuscation, or multi-stage strategies. In contrast, LLMs and their fine-tuned derivatives have demonstrated a remarkable ability to understand, generate, and contextualize human language, making them uniquely suited for analysing unstructured data. This paper advances the existing literature by offering a holistic framework that categorizes LLM applications into six key cyber security domains, including vulnerability detection, anomaly detection, cyber threat intelligence, blockchain security, penetration testing, and digital forensics. These domains were selected based on the maturity of research, practical impact, and the demonstrated value of LLMs in addressing real-world cyber security challenges. Within each domain, this paper examines in depth how LLMs contribute to context-aware reasoning, automation, and adaptability to emerging threats (Figure 3).

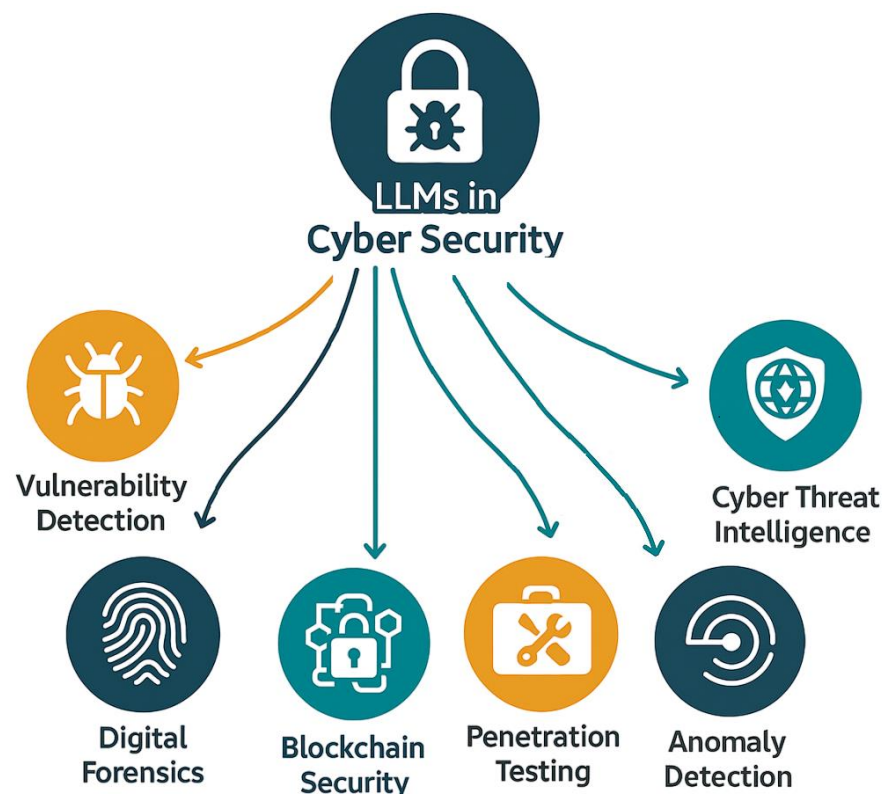


Figure 3. Holistic framework to categorize LLM applications into six key cyber security domains.

This categorization not only brings clarity to the fragmented landscape of LLM applications in cyber security but also serves as a structured lens through which researchers and practitioners can assess current capabilities and identify future opportunities. The chart, shown in Figure 4, reinforces this categorization by illustrating the distribution of published research papers across the six identified domains from 2023 to 2025. Notably, vulnerability detection (VD) and anomaly detection (AD) exhibit steady growth, reflecting their foundational role in proactive cyber defence. Cyber threat intelligence (CTI) saw a dramatic spike in 2024, suggesting heightened research interest in leveraging LLMs for extracting actionable insights from threat data. In contrast, areas like penetration testing (Pentest) and blockchain security (BC) show relatively modest and consistent publication volumes, potentially indicating emerging research interest in these areas. Digital forensics (DF), while initially underexplored in 2023, witnessed significant attention in 2024, possibly driven by new investigative use cases and tools.

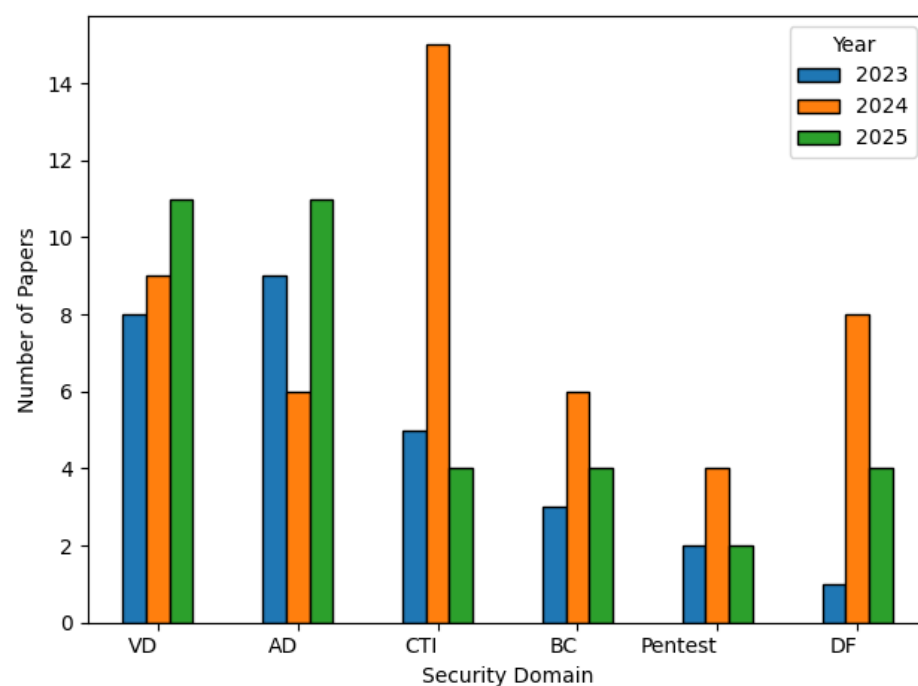


Figure 4. Published papers on LLM-based systems in each cyber security domain (2023–2025).

4.1. Vulnerability Detection

The application of LLMs in vulnerability detection has rapidly evolved into one of the most active and impactful areas of LLM-driven cyber security research. A wide range of studies reviewed by Zhang et al. [42] have explored how LLMs can enhance traditional vulnerability detection techniques by improving automation, reducing false positives, and enabling the semantic understanding of code and configurations. The key to this evolution is the integration of LLMs into both static and dynamic analysis pipelines.

Several studies have demonstrated the use of LLMs to detect vulnerabilities directly from source code. Zhou et al. [43,44], and Tamberg and Bahsi [45] proposed approaches using LLMs to identify insecure coding patterns and semantic weaknesses, emphasizing the benefit of pretrained language models over traditional pattern-matching tools. Similarly, Mahyari [46] and Mao et al. [47] leveraged LLM models, like CodeBERT and GPT-3, for vulnerability classification tasks, showing that LLMs could generalize across different programming languages and vulnerability types with minimal retraining. Beyond classification, several researchers focused on vulnerability localization and explanation. For example, Cheshkov et al. [48] designed a method to highlight precise code lines contribut-

ing to a vulnerability, while Purba et al. [49] applied LLMs to reason about causality in code defects. These efforts represent a shift toward interpretable vulnerability detection, where models not only identify vulnerable code but also offer human-readable explanations.

Dozono et al. [50] extended this evaluation across multiple programming languages, developing CODEGUARDIAN, a VSCode-integrated tool that enhances developers' accuracy and speed in detecting vulnerabilities. Berabi et al. [51] also proposed DeepCode AI Fix, a system that utilizes program analysis to focus LLMs' attention on relevant code snippets, thereby improving the efficiency of vulnerability fixes. Similarly, Noever [37] assessed GPT-4's performance against traditional static analysers, like Snyk and Fortify, finding that GPT-4 identified approximately four times more vulnerabilities and provided viable fixes with a low false positive rate. Sheng et al. [52] also proposed LProtector, an LLM-driven vulnerability detection system that leverages GPT-4o and Retrieval-Augmented Generation (RAG) to identify vulnerabilities in C/C++ codebases, outperforming State-of-the-Art baselines in F1 score.

LLMs have also been integrated into software development environments. Khare et al. [53] and Jensen et al. [54] demonstrated how developer-facing tools powered by LLMs could offer real-time feedback on insecure code during the writing process, bridging the gap between detection and secure development. Shestov et al. [55] extended this by proposing LLM-assisted bug reporting systems that correlate vulnerabilities with developer comments and test results. Other works, like those by Li et al. [56] and Kouliaridis et al. [2], explored the fine-tuning of LLMs for vulnerability datasets, improving detection precision and recall. Guo et al. [57] and Wang et al. [58] focused on model robustness in the face of obfuscated or adversarial code, an important area given the tendency of attackers to hide exploit logic. At the infrastructure level, Bakhshandeh et al. [59] and Mathews et al. [60] showed how LLMs could identify misconfigurations in containerized environments and cloud infrastructures, extending vulnerability detection beyond traditional application code. LLMs were also utilized as virtual security response experts. Lin et al. [61] demonstrated that LLMs could assist in vulnerability mitigation by retrieving relevant CVE and CWE data and suggesting remediation strategies. Kaheh et al. [62] took this a step further, arguing that LLMs should not only answer queries but also perform actions, such as instructing a firewall to block a malicious IP.

Recent research has increasingly focused on the application of LLMs for vulnerability detection and code security. Dolcetti et al. [63] and Feng et al. [64] explored enhancing LLM performance by integrating them with traditional techniques like testing, static analysis, and code graph structures, improving both detection and repair capabilities. Similarly, Lin and Mohaisen [65] investigated the effectiveness of fine-tuned LLMs and general-purpose models, highlighting the role of dataset quality and token length in shaping model reliability. Torkamani et al. [66] proposed CASEY, a triaging tool that accurately classifies CWEs and predicts severity levels, demonstrating practical utility in automating security assessments. In parallel, broader methodological and empirical advancements have surfaced. Ghosh et al. [67] introduced CVE-LLM, embedding ontology-driven semantics from CVE and CWE databases to enhance context-aware vulnerability evaluations. Also, Sheng et al. [7] provided a State-of-the-Art survey mapping out the LLM landscape in software security and identifying key challenges. Sovrano et al. [39] also offered critical insight into LLM limitations by identifying the "lost-in-the-end" effect, where vulnerabilities near the input's end are frequently missed, calling for smarter input management strategies. Table 2 summarizes the contributions and limitations of recent LLM-based systems for vulnerability detection in ascending order per year of publication.

Table 2. Summary of contributions and limitations of recent LLM-based systems for vulnerability detection.

Citation	Year	Summary of Contributions	Limitations
Cheshkov et al. [48]	2023	Designed LLMs for vulnerability localization at the line level and provided explanation support for detected issues.	Accuracy decreases with obfuscated code or multiple bugs.
Purba et al. [49]	2023	Introduced causality-aware vulnerability analysis using LLMs, which enabled deeper reasoning about code defect origins.	Computationally intensive and less scalable for large codebases.
Noever [37]	2023	Assessed GPT-4 against traditional analysers and identified more vulnerabilities with viable fixes.	Risk of over-prediction and requires expert validation.
Li et al. [56]	2023	Fine-tuned LLMs on labelled vulnerability datasets for higher detection accuracy.	Requires well-labelled, balanced datasets to avoid bias.
Wang et al. [58]	2023	Combined code embeddings with LLMs for better generalization across codebases.	Requires large compute resources and careful tuning.
Bakhshandeh et al. [59]	2023	Used LLMs to detect configuration-related vulnerabilities in Docker and Kubernetes environments.	Limited support for custom or evolving infrastructure setups.
Lin et al. [61]	2023	Demonstrated LLMs' capability to assist with vulnerability analysis by retrieving and explaining CVEs and CWEs.	May retrieve outdated or irrelevant vulnerabilities in noisy contexts.
Kaheh et al. [62]	2023	Proposed interactive LLMs that go beyond Q&A to take actions, like configuring defences.	Unverified automation can endanger production systems.
Zhou et al. [43,44]	2024	Proposed LLM-based models to detect insecure code patterns in source files.	Limited generalization across codebases.
Mahyari [46]	2024	Fine-tuned LLMs to classify various vulnerability types in codebases and then emphasized cross-language generalization.	Performance drops with rare or unseen vulnerability types.
Khare et al. [53]	2024	Developed IDE-integrated LLM tools to offer live vulnerability feedback to developers.	Requires significant work for IDE compatibility and user trust.
Jensen et al. [54]	2024	Proposed real-time LLM-based advisors in development environments.	Susceptible to hallucinations or incomplete recommendations.
Shestov et al. [55]	2024	Introduced bug reporting that integrates developer context with LLM-generated vulnerability, which enhances report clarity.	Performance suffers when the developer context is noisy or unavailable.
Mathews et al. [60]	2024	Proposed cloud-specific LLM models to assess misconfiguration risks in cloud setups.	May not generalize well across different cloud platforms.
Berabi et al. [51]	2024	Proposed DeepCode AI Fix, focusing LLMs on relevant code snippets to improve fix efficiency.	Limited to codebases with clear function or method structures.

Table 2. Cont.

Citation	Year	Summary of Contributions	Limitations
Dozono et al. [50]	2024	Developed CODEGUARDIAN, enhancing multi-language vulnerability detection efficiency.	Performance varies by language and toolchain integration.
Kouliaridis et al. [2]	2025	Improved vulnerability classification and filtering by aligning LLM outputs with CVE taxonomies.	Mapping errors can occur in multi-class or ambiguous cases.
Sheng et al. [52]	2025	Proposed LProtector, leveraging GPT-4o and RAG for effective vulnerability detection in C/C++ codebases.	High computational cost and requires large context windows for best results.
Tamberg & Bahsi [45]	2025	Used transformer models to detect semantic-level vulnerabilities.	Heavily reliant on the quality of prompts and pretraining data.
Mao et al. [47]	2025	Investigated GPT-based vulnerability scanners for C/C++ code. Improved classification precision using minimal supervised data.	Struggles with complex or deeply nested code structures.
Ghosh et al. [67]	2025	Proposed CVE-LLM using ontologies and LLMs to automate vulnerability evaluation.	Relies on structured ontologies and struggles with novel CVEs.
Sheng et al. [7]	2025	Provided a review of LLM-based vulnerability detection, mapping techniques and future trends.	Lacks empirical validation.
Sovrano et al. [39]	2025	Presented LLMs' missed vulnerabilities at the end of large files ("lost-in-the-end" effect).	The proposed model suffers from location-based bias.
Torkamani et al. [66]	2025	Proposed CASEY, an LLM system for CWE and severity classification in vulnerability triage.	Combined classification is challenging.
Lin & Mohaisen [65]	2025	Evaluated LLMs on Java vulnerability detection and highlight token length sensitivity.	Performance varies widely and is affected by input truncation
Dolcetti et al. [63]	2025	Combined LLMs with testing and static analysis to auto-correct flawed code generations.	Not robust in unsupervised settings.
Feng et al. [64]	2025	Proposed CGP-Tuning to integrate code graph structures into LLM prompt tuning.	Do not generalize across all programming languages.

Although research on LLM-based vulnerability detection continues to advance, it also uncovers recurring challenges that limit practical deployment. Many systems, such as those proposed by Li et al. [56] and Mahyari [46], rely on fine-tuning LLMs with labelled datasets to enhance accuracy. However, this approach introduces bias risks and reduces performance on rare or unseen vulnerabilities. Several works, including those by Purba et al. [49] and Wang et al. [58], focus on enriching LLM reasoning or generalization through causality analysis or code embeddings but at the cost of scalability and compute intensity. A common strength across reviewed studies, such as those by Khare et al. [53], Jensen et al. [54], and Shestov et al. [55], is integration with developer environments for real-time feedback and contextualized insights. However, these systems suffer from trust issues, hallucinations, or reliance on high-quality developer context. Additionally, limitations

in generalizability across codebases or languages [38,45] and challenges in handling obfuscated or deeply nested code [42,43] highlight persistent gaps. Other studies aim to address these weaknesses by ensuring scalability, cross-language robustness, and enhanced resilience to code complexity, which is lacking in the current systems [61,62].

Several LLM-based tools show promising potential for production deployment. CODEGUARDIAN [50], for example, is integrated into Visual Studio Code, enabling real-time vulnerability detection during coding, which is a valuable feature for secure software development pipelines. DeepCode AI Fix [51] extends this by not only identifying but also suggesting fixes, making it suitable for continuous integration workflows. LProtector [52] combines GPT-4o with RAG to effectively detect C/C++ vulnerabilities in large codebases, showing promise for enterprise-scale applications. These tools go beyond experimental use and are designed with real-world developer workflows in mind. Their architecture and integration features position them well for practical deployment; however, broader industry validation is still needed.

4.2. Anomaly Detection

Anomaly detection plays a crucial role in cyber security, and recent studies have increasingly leveraged LLMs to enhance this capability across various contexts, from system logs to email filtering and traffic inspection. A standout contribution in log-based anomaly detection is LogPrompt [68], which utilizes LLMs to parse and semantically understand unstructured logs. It applies chain-of-thought reasoning and in-context learning to distinguish between normal and anomalous behaviour, thus moving beyond static rule sets. LEMUR [69] further expands this field by employing entropy sampling for efficient log clustering and uses LLMs to differentiate between invariant tokens and parameters with high accuracy, greatly improving the effectiveness of template merging. In a practical cloud environment, ScaleAD [70] offers an adaptive anomaly detection system where a Trie-based agent flags potential anomalies, and an LLM is queried to assess log semantics and assign confidence scores. Similarly, LogGPT [71] introduces a full pipeline that preprocesses logs, crafts context-rich prompts, and parses LLM outputs for anomaly evaluation. These techniques allow the system to intelligently detect anomalies even in high-volume environments. Adding to the landscape, Han et al. [72] improve GPT-2's focus in log sequence analysis through a Top-K reward mechanism, which directs model attention to the most relevant tokens, thus enhancing detection precision.

Outside of logs, LLMs have been evaluated for their performance in phishing and spam detection. The Improved Phishing and Spam Detection Model (IPSDM) [73] fine-tunes DistilBERT and RoBERTa for email filtering, showing LLMs' promise in real-time protection against social engineering. Si et al. [74] also highlight ChatGPT's potential, outperforming BERT on low-resource Chinese datasets but showing limitations on large-scale English corpora. Nahmias et al. [38] targeted spear-phishing detection with a unique approach using LLMs to generate contextual document vectors that reflect persuasion tactics, outperforming classical classifiers in recognizing malicious intent. Further, Heiding et al. [75] examined the dual-use nature of LLMs in both crafting and detecting phishing emails. Their analysis demonstrates that models like GPT-4, Claude, and PaLM outperform humans in detection tasks, though they warn of adversarial misuse. An additional threat domain is covered by Vörös et al. [76], who propose a student-teacher LLM approach for malicious URL detection via knowledge distillation, making the model suitable for lightweight deployments. Also, Guastalla et al. [77] fine-tuned LLMs for DDoS detection on pcap-based and IoT datasets, adapting to real-world traffic complexities.

Other research has extended LLM-based anomaly detection beyond logs and text into diverse data modalities. Alnegheimish et al. [78] proposed sigllm, a zero-shot time series anomaly detector that converts numerical sequences into a textual form, enabling LLMs to forecast and flag anomalies without specific training. Gu et al. [79] introduced AnomalyGPT, leveraging vision–language models like MiniGPT-4 for industrial defect detection using synthetic image–text pairs, eliminating manual thresholding. Elhafsi et al. [80] also applied LLMs in robotic systems, identifying semantic inconsistencies in visual behaviours, while Li et al. [81] demonstrated that LLMs can detect outliers in tabular data with zero-shot capabilities, which was further improved through fine-tuning for real-world contexts.

Guan et al. [82] proposed a framework combining BERT and LLaMA models to detect anomalies in system records. By aligning semantic vectors from log messages, the model effectively captures anomalies without relying on traditional log parsing methods. Su et al. [83] conducted a systematic literature review on the use of LLMs in forecasting and anomaly detection. The review highlights the potential of LLMs in these areas while also addressing challenges such as data requirements, generalizability, and computational resources. Ali and Kostakos [84] integrated LLMs with traditional ML techniques and explainability tools, like SHAP, in a hybrid model called HuntGPT, which enriches anomaly detection through human-interpretable insights.

Recently, Liu et al. [85] introduced CYLENS, an LLM-powered assistant that supports security analysts in interpreting anomalies, attributing threats, and suggesting mitigations by leveraging contextual memory and historical CTI through expert-style dialogues. Wang and Yang [86] also proposed a federated LLM framework that performs anomaly detection across distributed and privacy-sensitive environments by processing multimodal data without centralizing it. Qian et al. [87] focused on Android malware detection with LAMD, a multi-tiered system that integrates program slicing, control flow graphs, and LLM prompting to detect and explain malicious behaviour while also reducing hallucinations through consistency verification. Benabderrahmane et al. [88] target stealthy APTs with APT-LLM, encoding low-level system traces into text and using LLMs to generate semantic embeddings, which are analysed via autoencoders to detect subtle deviations.

Additionally, Meguro and Chong [89] proposed AdaPhish to advance phishing defence by automating phishing email analysis and anonymization using LLMs and vector databases, which serve both detection and organizational education. Another tool named SHIELD [4] goes beyond detection by combining statistical and graph-based techniques with LLM-generated APT kill chain narratives, which provide security analysts with actionable and human-readable summaries for stealthy threat detection. In the same way, Akhtar et al. [90] reviewed the LLM-log analysis landscape and identified major methodologies, like fine-tuning, RAG, and in-context learning, and outlined the trade-offs and practical challenges in real deployments. Walton et al. [91] also introduced MSP, which enhances malware reverse engineering with structured LLM-driven summarization that categorizes threats and pinpoints specific malicious behaviours. Table 3 provides a summary of the contributions and limitations of recent LLM-based systems for anomaly detection in ascending order per year of publication.

Table 3. Summary of contributions and limitations of recent LLM-based systems for anomaly detection.

Citation	Year	Summary of Contributions	Limitations
Liu et al. [70]	2023	Proposed ScaleAD, combining Trie-based detection with LLM validation for cloud logs.	LLM query latency may limit real-time responses.
Qi et al. [71]	2023	Proposed LogGPT, a modular framework for LLM-based log anomaly detection.	Depends on the prompt design and parsing accuracy.
Han et al. [72]	2023	Enhanced GPT-2 with a Top-K reward strategy for focused log anomaly detection.	Task-specific tuning is required for each log type.
Jamal & Wimmer [73]	2023	Built IPSDM, a phishing/spam detector fine-tuned on RoBERTa and DistilBERT.	It underperforms on novel email attack formats.
Heiding et al. [75]	2023	Evaluated GPT-4 for generating and detecting phishing emails, surpassing traditional methods.	Dual-use risks and phishing generation could be abused.
Vörös et al. [76]	2023	Used knowledge distillation to detect malicious URLs with lightweight student models.	Performance varies based on teacher model quality.
Gu et al. [79]	2023	Proposed AnomalyGPT for industrial anomaly detection using LVLMS.	Requires synthetic data generation and fine-tuning.
Elhafsi et al. [80]	2023	Proposed LLM-based monitoring for semantic anomalies in robotics.	Focused only on specific robotic applications.
Ali & Kostakos [84]	2023	Proposed HuntGPT, integrating LLMs with XAI and ML for enhanced anomaly explainability.	Dependent on model interpretability tools.
Liu et al. [68]	2024	Proposed LogPrompt, a log analysis method using LLMs with CoT reasoning and in-context learning.	Requires quality training data and struggles with noisy logs.
Nahmias et al. [38]	2024	Developed a spear-phishing detector using LLM-generated contextual vectors.	Requires LLM access for inference and vector interpretation.
Guastalla et al. [77]	2024	Applied LLMs to DDoS detection on cloud and IoT datasets using few-shot learning.	Scalability issues for massive traffic volumes.
Alnegheimish et al. [78]	2024	Introduced sigllm for zero-shot time series anomaly detection using LLMs.	Performance lags behind specialized DL models.
Li et al. [81]	2024	Demonstrated zero-shot anomaly detection in tabular data using LLMs.	Effectiveness depends on data representation quality.
Su et al. [83]	2024	Provided a comprehensive review of LLMs in forecasting and anomaly detection.	Lack of empirical results on data and computational costs.
Gandhi et al. [4]	2025	Developed hybrid APT detection using graphs, statistical models, and LLMs to explain alerts in the context of kill chains.	Requires careful graph tuning and high-quality logs for semantic consistency.
Zhang et al. [69]	2025	Proposed LEMUR, which enhances log clustering and template merging through semantic analysis.	High computational cost for entropy sampling.

Table 3. Cont.

Citation	Year	Summary of Contributions	Limitations
Si et al. [74]	2025	Evaluated ChatGPT on spam detection, showing strength in low-resource contexts.	Performance degrades on complex and large-scale datasets.
Guan et al. [82]	2025	Proposed LogLLM for detecting anomalies in system logs using LLMs.	Requires adaptation for different log formats.
Liu et al. [85]	2025	Developed an LLM-powered assistant for anomaly interpretation and mitigation support using CTI-enhanced dialogue and prompting.	Requires extensive fine-tuning and context modelling.
Wang & Yang [86]	2025	Proposed a federated LLM framework for privacy-preserving, multimodal anomaly detection across distributed systems.	A federated setup may increase system complexity and latency.
Qian et al. [87]	2025	Designed a multi-tier LLM-based Android malware detection framework combining static slicing, CFGs, and hallucination filters.	Limited to Android apps and affected by obfuscation techniques.
Benabderrahmane et al. [88]	2025	Proposed APT-LLM, which uses LLM-based embeddings of provenance traces combined with autoencoders to detect APTs on multiple OS.	High computational cost and generalization remain challenges for practical use.
Meguro & Chong [89]	2025	Developed a real-time phishing detection tool using LLMs with vector database support.	It is based on email formats and needs large historical datasets.
Akhtar et al. [90]	2025	Reviewed LLMs in log analysis for anomaly detection, root cause analysis, and threat tracing.	Lack of empirical results.
Walton et al. [91]	2025	Introduced LLM-based semantic processor for Android malware classification using GPT-4o-mini.	It depends on prompt quality and obfuscation techniques.

The reviewed literature demonstrates varied approaches to utilize LLMs in anomaly detection but also highlights common limitations. Several studies, such as those by Qi et al. [71] and Liu et al. [70], focus on LLM-based log analysis, emphasizing structured outputs and integration with existing systems. However, these approaches suffer from latency issues or reliance on prompt quality. Similarly, phishing and spam detection systems [70,73] showcase the effectiveness of LLMs in recognizing persuasive language but are hindered by risks of misuse, generalization gaps, or poor performance on novel formats. Other studies proposed solutions to address issues of real-time detection, cross-domain generalization, and reduced dependency on handcrafted prompts. For example, systems, like those by Gandhi et al. [4] and Benabderrahmane et al. [88], attempt to provide deeper semantic understanding through kill chain alignment or provenance embeddings; however, they still face challenges in computational overhead and interpretability. While many papers suggest innovative techniques, like federated learning [86] or zero-shot detection [81], their practical deployment remains limited.

4.3. Cyber Threat Intelligence (CTI)

The integration of LLMs into CTI has become a prominent area of study, as researchers assess how these models can assist with threat detection, analysis, and decision-making.

Recent advances in CTI leverage LLMs to transform unstructured and distributed threat data into actionable insights. Mitra et al. [92] proposed LocalIntel, a framework that synthesizes global threat databases, like CVE and CWE, with organization-specific knowledge, allowing for personalized, high-context intelligence summaries. Perrina et al. [93] automate CTI report generation using LLMs to extract threat knowledge from unstructured sources, highlighting LLMs' capability in structuring scattered data. Fayyazi and Yang [94] also fine-tuned LLMs with structured threat ontologies, resulting in models that excel in producing precise, technique-specific threat narratives. Also, Schwartz et al. [95] proposed LMCloudHunter, which translates OSINT into candidate detection rules for signature-based systems using LLMs, thus connecting high-level intelligence with operational defences.

Other studies have focused on extracting CTI from unstructured text. For example, Clairoux-Trepanier et al. [96] evaluated GPT-based LLMs for extracting key threat indicators, showing promise in identifying IOCs and attack tactics. Siracusano et al. [97] introduced the aCTIon framework, combining LLMs with a parsing engine to export CTI in the STIX format, enabling integration with existing intelligence platforms. Hu et al. [98] extended this idea using fine-tuned LLMs to build and populate knowledge graphs from raw CTI, improving the structured representation. Zhang et al. [99] also proposed AttackKG, a full LLM-based pipeline that rewrites and summarizes CTI into attack graphs. Fieblinger et al. [100] validated open-source LLMs for extracting entity–relation triples, confirming their role in enriching semantic CTI knowledge graphs.

Park and You [101] proposed CTI-BERT, a domain-adapted BERT model trained on cyber security-specific corpora to enhance the extraction of structured threat intelligence from unstructured data. Gao et al. [102] also presented ThreatKG, which uses LLMs to extract entities and relationships from CTI reports to construct and update a cyber threat knowledge graph, supporting continuous integration into analytic workflows. To address the challenge of limited labelled data, Sorokoletova et al. [103] proposed 0-CTI, a versatile framework supporting both supervised and zero-shot learning for threat extraction and STIX alignment, ensuring standardized data communication. Wu et al. [104] focused on verifying extracted intelligence, using LLMs to identify key claims in reports and validate them against a structured knowledge base, enhancing the credibility of threat intelligence. Fayyazi et al. [105] demonstrated how integrating RAG with LLMs enriches and improves the accuracy of MITRE ATT&CK TTP summaries. Singla et al. [106] also showed LLMs' effectiveness in analysing software supply chain vulnerabilities and identifying causes and propagation paths. Zhang et al. [107] tackled bug report deduplication by enhancing traditional similarity-based models with LLM-driven semantic insights, leading to more accurate identification of duplicate reports.

To support strategic defence reasoning, Jin et al. [3] proposed Crimson, a system that aligns LLM outputs with MITRE ATT&CK tactics through a novel Retrieval-Aware Training (RAT) approach. This fine-tuning method significantly reduced hallucinations while improving alignment with expert knowledge. Similarly, Tseng et al. [108] proposed an LLM-powered agent to automate SOC tasks, such as extracting CTI from reports and generating regular expressions for detection rules. Rajapaksha et al. [109] also introduced a QA system using LLMs and RAG to support context-rich attack investigations. Shah and Parast [110] applied GPT-4o with one-shot fine-tuning to automate CTI workflows, reducing manual effort while ensuring accuracy. Shafee et al. [5] assessed LLM-driven chatbots for OSINT, showing their ability to classify threat content and extract entities. Alevizos and Dekker [111] also proposed an AI-powered CTI pipeline that fosters collaboration between analysts and LLMs, enhancing both speed and precision in threat intelligence generation. Daniel et al. [112] also evaluated LLMs' ability to map Snort NIDS rules to MITRE ATT&CK techniques, highlighting their scalability and explainability. Similarly,

Alturkistani et al. [113] explored the significant role of LLMs in ICT education, highlighting their potential to enhance personalised learning, automate content generation, and support practical skill development. Table 4 summarizes the contributions and limitations of recent LLM-based systems for CTI in ascending order per year of publication.

Table 4. Summary of the contribution and limitations of recent LLM-based systems in CTI.

Citation	Year	Summary of Contributions	Limitations
Hu et al. [98]	2022	Built knowledge graphs from threat data and used LLMs to populate them automatically with high-quality relationships.	LLMs inferred inaccurate relationships without grounding.
Perrina et al. [93]	2023	Developed an LLM-based system for extracting threat data from unstructured sources and automating CTI report generation.	Risk of hallucination and unverified outputs in generated reports.
Fayyazi & Yang [94]	2023	Fine-tuned LLMs using MITRE ATT&CK and CAPEC to improve the quality of threat technique descriptions.	Limited generalization outside training data.
Siracusano et al. [97]	2023	Proposed aCTIon, combining LLMs with structured parsing to export CTI in STIX format for integration into CTI platforms.	Errors can propagate during STIX export if parsing fails or the context is misread.
Park & You [101]	2023	Proposed CTI-BERT, a BERT-based model trained on cyber security data, enhancing CTI extraction accuracy over general models.	Restricted to domains represented in its fine-tuning data.
Singla et al. [106]	2023	Demonstrated LLMs can analyse supply chain vulnerabilities and identify threat propagation paths.	Lack of generalizability as it focused on specific software.
Jin et al. [3]	2024	Proposed Crimson, aligning LLM outputs with MITRE ATT&CK using RAT to reduce hallucinations.	Requires well-structured retrieval corpora to function effectively.
Shafee et al. [5]	2024	Evaluated LLM-based chatbots' performance in classifying and extracting entities from OSINT data, enhancing threat awareness.	Varying accuracy across LLMs, which are heavily dependent on prompt design.
Schwartz et al. [95]	2024	Proposed LMCloudHunter, which extracts detection rules from OSINT and translates them into signatures.	Needs manual validation to ensure safety of signatures.
Clairoux-Trepanier et al. [96]	2024	Evaluated LLMs for extracting IOCs from CTI reports, showing their strength in semantic understanding.	Limited effectiveness with inconsistent or noisy data formats.
Zhang et al. [99]	2024	Proposed a four-stage pipeline using LLMs to rewrite, parse, and summarize threat data into structured attack knowledge graphs.	High complexity, where each stage adds potential failure points.
Fieblinger et al. [100]	2024	Utilized LLMs to extract entity–relation triples from CTI, helping automated graph structure and analysis.	Performance varies depending on entity consistency and text clarity.

Table 4. Cont.

Citation	Year	Summary of Contributions	Limitations
Gao et al. [102]	2024	Proposed ThreatKG, an automated system constructing and updating a threat knowledge graph from open-source CTI reports.	Suffers from incomplete or inconsistent entity linking.
Wu et al. [104]	2024	Proposed an LLM-based verification method to advance the credibility of unverified threat intelligence.	Requires robust and updated knowledge bases for validation.
Fayyazi et al. [105]	2024	Enhanced TTP summarization with LLMs using RAG, improving factual accuracy and depth.	Dependent on retrieval source quality and relevance.
Zhang et al. [107]	2024	Improved bug report deduplication using semantic insights from LLMs, outperforming traditional models.	Sensitive to wording variations.
Tseng et al. [108]	2024	Proposed an LLM agent to automate SOC tasks, including regex generation and CTI extraction from threat reports.	Prone to generating overly generic or unsafe regular expressions.
Rajapaksha et al. [109]	2024	Developed a RAG-powered LLM QA system for attack search, providing context-aware help to SOC analysts.	Limited responsiveness when external retrieval fails or is slow.
Shah & Parast [110]	2024	Explored GPT-4o and one-shot fine-tuning to automate CTI processes, reducing manual effort.	Lacks transparency in generated outputs.
Alevizos & Dekker [111]	2024	Proposed an AI-enhanced CTI processing pipeline, emphasizing AI-human collaboration for timely and high-fidelity threat intelligence.	Pipeline complexity may delay adoption in time-sensitive scenarios.
Daniel et al. [112]	2024	Evaluated LLMs (ChatGPT, Claude, and Gemini) for labelling NIDS rules with MITRE ATT&CK techniques, highlighting their scalability and explainability.	LLMs underperform compared to traditional ML models in accuracy metrics.
Mitra et al. [92]	2025	Proposed LocalIntel, combining global and local knowledge for tailored threat intelligence using LLMs.	Dependent on the availability and quality of local threat data.
Sorokoletova et al. [103]	2025	Proposed 0-CTI, a framework supporting supervised and zero-shot learning for CTI information extraction, aligning outputs with STIX.	Zero-shot performance declines sharply with vague or ambiguous language.
Liu et al. [85]	2025	Proposed CYLENS, an LLM-powered assistant supporting various stages of threat management by integrating extensive threat report knowledge.	Struggles to handle conflicting or overlapping intelligence from different sources.
Paul et al. [114]	2025	Combined LLMs with RAG to process real-time threat intelligence feeds, enabling dynamic and context-aware threat detection.	Dependent on the quality and freshness of streaming threat feeds.

The reviewed studies highlight the growing role of LLMs in enhancing CTI through automation, semantic understanding, and integration with structured formats. Many studies, such as those by Perrina et al. [93], Siracusano et al. [97], and Zhang et al. [99],

employ LLMs to convert unstructured threat data into structured formats, like STIX or knowledge graphs, offering semantic depth and automation. However, these systems suffer from limitations like hallucinated outputs, parsing errors, and cumulative complexity across multi-stage pipelines. Similarly, RAG-based approaches, such as those by Jin et al. [3] and Fayyazi et al. [105], improve factual accuracy by grounding responses in external sources, but their performance heavily depends on the quality and structure of the retrieval corpus. Tools designed for operational use, such as those by Wu et al. [104] and Rajapaksha et al. [109], show promise in supporting SOC analysts through verification and context-aware reasoning; however, they struggle with latency and incomplete context. Furthermore, domain-specific models like CTI-BERT [101] offer improved accuracy but lack generalizability. A common limitation across various studies, especially those by Hu et al. [98] and Liu et al. [85], is the difficulty in grounding LLM-generated outputs and managing conflicting or ambiguous intelligence, which poses risks to reliability and practical deployment.

4.4. Blockchain Security

Recent advancements in blockchain security have seen the integration of LLMs to enhance smart contract auditing and anomaly detection. Chen et al. [115] conducted an empirical study assessing ChatGPT's performance in identifying vulnerabilities within smart contracts. Their findings indicate that while ChatGPT exhibits high recall, its precision is limited, with performance varying across different vulnerability types. David et al. [116] also explored the feasibility of employing LLMs, specifically GPT-4 and Claude, for smart contract security audits. Their research focused on optimizing prompt engineering and evaluating model performance using a dataset of 52 compromised DeFi smart contracts. The study revealed that LLMs correctly identified vulnerability types in 40% of cases but also demonstrated a high false positive rate, underscoring the continued necessity for manual auditors.

Gai et al. [117] proposed BLOCKGPT, a dynamic, real-time approach to detecting anomalous blockchain transactions. By training a custom LLM from scratch, BLOCKGPT acts as an IDS, effectively identifying abnormal transactions without relying on predefined rules. In experiments, it successfully ranked 49 out of 124 attacks among the top three most abnormal transactions, showcasing its potential in real-time anomaly detection. Hu et al. [118] proposed GPTLENS, an adversarial framework that enhances smart contract vulnerability detection by breaking the process into two stages: generation and discrimination. In this framework, the LLM serves dual roles as both AUDITOR and CRITIC, aiming to balance the detection of true vulnerabilities while minimizing false positives. The experimental results demonstrate that this two-stage approach yields significant improvements over conventional one-stage detection methods. Similarly, Sun et al. [119] proposed GPTScan, a tool that combines GPT with static analysis to detect logic vulnerabilities in smart contracts. Unlike traditional tools that focus on fixed-control or data-flow patterns, GPTScan leverages GPT's code understanding capabilities to identify a broader range of vulnerabilities. The evaluations on diverse datasets showed that GPTScan achieved high precision for token contracts and effectively detected ground-truth logic vulnerabilities, including some missed by human auditors. Hossain et al. [120] combined LLMs with traditional ML models to detect vulnerabilities in smart contract code, achieving over 90% accuracy.

Yu et al. [121] proposed Smart-LLaMA, a vulnerability detection framework leveraging domain-specific pretraining and explanation-guided fine-tuning to identify and explain smart contract flaws with improved accuracy. Similarly, Zaazaa and El Bakkali [122] proposed SmartLLMSentry, which employs ChatGPT and in-context learning to detect

Solidity-based vulnerabilities with over 90% precision. Also, Wei et al. [123] proposed LLM-SmartAudit, a multi-agent conversational system that enhances auditing by outperforming traditional tools in identifying complex logical issues. He et al. [124] conducted a comprehensive review on integrating LLMs in blockchain security tasks, covering smart contract auditing, anomaly detection, and vulnerability repair. Moreover, Ding et al. [125] proposed SmartGuard, which uses LLMs to retrieve semantically similar code snippets and apply Chain-of-Thought prompting to improve smart contract vulnerability detection, achieving near-perfect recall. Ma et al. [126] proposed iAudit, a two-stage LLM-based framework for smart contract auditing. The system employs a Detector to flag potential vulnerabilities and a Reasoner to provide contextual explanations, improving audit interpretability. However, its effectiveness is contingent on the quality and diversity of fine-tuning data. Similarly, Bu et al. [127] introduced vulnerability detection in DApps by fine-tuning LLMs on a large dataset of real-world smart contracts. Their method excelled in identifying complex, non-machine-auditable issues, achieving high precision and recall. However, its robustness depends on the continual inclusion of diverse, evolving contract patterns. Table 5 summarizes the contributions and limitations of recent LLM-based systems in blockchain security in ascending order per year of publication.

Table 5. Summary of the contribution and limitations of recent LLM-based systems in blockchain security.

Citation	Year	Summary of Contributions	Limitations
David et al. [116]	2023	Evaluated GPT-4 and Claude for smart contract audits, achieving a 40% correct identification rate.	A high false positive rate necessitates manual auditing.
Gai et al. [117]	2023	Proposed BLOCKGPT for real-time anomaly detection in blockchain transactions using a pretrained LLM.	Effectiveness relies on high-quality, extensive training data.
Hu et al. [118]	2023	Proposed GPTLENS, a two-stage LLM framework enhancing vulnerability detection in smart contracts by separating generation and discrimination.	Complexity in implementation and challenges in balancing detection and false positives.
Sun et al. [119]	2024	Proposed GPTScan, combining GPT with static analysis to detect vulnerabilities in smart contracts.	Effectiveness relies on the quality of the static analysis.
Yu et al. [121]	2024	Proposed Smart-LLaMA using domain-specific training to detect and explain vulnerabilities.	Requires large and annotated datasets for optimal performance.
Zaazaa and El Bakkali [122]	2024	Proposed SmartLLMSentry using ChatGPT and in-context learning for vulnerability detection.	GPT-4 was less effective at rule generation than GPT-3.
Wei et al. [123]	2024	Proposed LLM-SmartAudit, a multi-agent system to enhance smart contract auditing.	System complexity increases with multiple AI agents.
Ma et al. [126]	2024	Proposed iAudit with a two-stage fine-tuning approach for detection and explanation.	Relies heavily on the quality of fine-tuning data.
Chen et al. [115]	2024	Evaluated ChatGPT's ability to detect smart contract vulnerabilities, noting high recall but limited precision.	Performance varies across vulnerability types.

Table 5. Cont.

Citation	Year	Summary of Contributions	Limitations
Hossain et al. [120]	2025	Combined LLMs with ML models to detect smart contract vulnerabilities, achieving high accuracy.	Limited validation on newer blockchain platforms.
He et al. [124]	2025	Reviewed LLM use in blockchain security, covering auditing, anomaly detection, and repair.	Lack of empirical results.
Ding et al. [125]	2025	Proposed SmartGuard, combining CoT prompting with LLMs for high-recall detection.	Less effective when multiple vulnerabilities coexist in code.
Bu et al. [127]	2025	Enhanced vulnerability detection in DApps by fine-tuning LLMs on a large dataset of smart contracts.	Require significant computational resources for fine-tuning.

The reviewed studies highlight the growing interest in applying LLMs for smart contract auditing and blockchain security, with varying approaches in model architecture, data usage, and integration techniques. A common trend among various studies, such as those by Hu et al. [118], Ma et al. [126], and Zaazaa and El Bakkali [126], is the adoption of multi-stage or fine-tuned frameworks aimed at improving vulnerability detection accuracy and interpretability. Other studies, including those by David et al. [116] and Chen et al. [115], focused on evaluating general-purpose LLMs, like GPT-4, revealing promising recall rates but also significant limitations in precision and high false positive rates. Real-time detection systems, like BLOCKGPT [117] and Smart-LLaMA [121], highlight the importance of domain-specific training; however, they suffer from a dependence on large, high-quality datasets. Similarly, systems like LLM-SmartAudit [123] and the one studied by Hossain et al. [120] explore hybrid architectures, though they introduce additional complexity and face challenges with generalization to evolving platforms. While tools like SmartGuard [125] and GPTScan [119] explore novel prompting or static analysis integration, their performance is constrained when multiple or subtle vulnerabilities exist. Generally, most systems demonstrate potential but struggle with generalizability, data dependency, scalability, or interpretability, highlighting the need for balanced, robust, and empirically validated solutions.

Several tools show growing readiness for real-world deployment, particularly in smart contract auditing. iAudit [126] simulates a structured audit process using LLM-driven detection and reasoning, making it suitable for automated contract security assessments. SmartLLMSentry [122] and Smart-LLaMA [121] apply pretrained and fine-tuned LLMs to detect vulnerabilities in Solidity code, achieving high accuracy on real-world datasets. GPTScan [119] and SmartGuard [125] combine static analysis with LLM reasoning to uncover logic vulnerabilities often overlooked by traditional tools. While large-scale production deployments are not yet widely documented, these tools demonstrate strong potential for integration into blockchain security workflows.

4.5. Penetration Testing

The integration of LLMs into penetration testing has seen significant advancements through recent research. Deng et al. [128] introduced PentestGPT, an automated framework that utilizes LLMs to perform tasks such as tool usage, output interpretation, and suggesting subsequent actions. Despite its modular design aimed at mitigating context loss, the study highlighted challenges in maintaining a comprehensive understanding throughout the testing process. Building upon this, Isozaki et al. [40] developed a benchmark to evaluate

LLMs like GPT-4o and Llama 3.1-405B in automated penetration testing. Their findings indicate that while these models show promise, they currently fall short in executing end-to-end penetration tests without human intervention, particularly in areas like enumeration, exploitation, and privilege escalation.

Exploring the offensive capabilities of LLMs, Fang et al. [129] demonstrated that GPT-4 could autonomously exploit 87% of tested one-day vulnerabilities when provided with CVE descriptions, outperforming other models and traditional tools. However, the model's effectiveness significantly decreased without detailed descriptions, exploiting only 7% of vulnerabilities. In the reconnaissance phase, Temara [130] examined ChatGPT's ability to gather information such as IP addresses and network topology, aiding in penetration test planning. While the tool provided valuable insights, the study emphasized the need for further validation of the accuracy and reliability of the data obtained. Deng et al. [131] evaluated PentestGPT, demonstrating a 228.6% increase in task completion compared to GPT-3.5. The modular design improved automation for penetration testing sub-tasks, though challenges with context-maintenance and complex scenarios persist.

Happe et al. [41] proposed a fully automated privilege escalation tool to evaluate LLMs in ethical hacking, revealing that GPT-4-turbo exploited 33–83% of tested vulnerabilities, outperforming GPT-3.5 and LLaMA3. The study also explored how factors like context length and in-context learning impact performance. Happe and Cito [132] also developed an AI sparring partner using GPT-3.5 to analyse machine states, recommend concrete attack vectors, and automatically execute them in virtual environments. Bianou et al. [133] proposed PENTEST-AI, a multi-agent LLM-based framework that automates penetration testing by coordinating agents through the MITRE ATT&CK framework. While it improves test coverage and task distribution, its reliance on predefined attack patterns limits adaptability to emerging threats. Table 6 summarizes the contributions and limitations of recent LLM-based systems in penetration testing in ascending order per year of publication.

Table 6. Summary of the contribution and limitations of recent LLM-based systems in penetration testing.

Citation	Year	Summary of Contributions	Limitations
Temara [130]	2023	Explored ChatGPT's utility in the reconnaissance phase of penetration testing.	Requires further validation of data accuracy and reliability.
Happe and Cito [132]	2023	Demonstrated LLMs as AI sparring partners for penetration testers, analysing machine states and suggesting attack vectors.	Requires further improvements and ethical considerations for deployment.
Deng et al. [128]	2024	Proposed an LLM-based framework for automating penetration testing tasks.	Challenges in maintaining context across tasks.
Fang et al. [129]	2024	Demonstrated GPT-4's ability to exploit one-day vulnerabilities with CVE descriptions.	Effectiveness diminishes without vulnerability descriptions.
Deng et al. [131]	2024	Evaluated PentestGPT's modular design and task completion improvements.	Struggles with complex scenarios and context retention.
Bianou et al. [133]	2024	Proposed PENTEST-AI, a multi-agent LLM framework integrating MITRE ATT&CK.	Reliance on predefined patterns limits adaptability to new threats.

Table 6. Cont.

Citation	Year	Summary of Contributions	Limitations
Isozaki et al. [40]	2025	Proposed a benchmark for evaluating LLMs in automated penetration testing.	LLMs struggle with end-to-end testing without human assistance.
Happe et al. [41]	2025	Developed an automated tool to assess LLMs' capability in privilege escalation tasks.	Struggles to maintain focus and manage errors in testing.

The reviewed studies highlight the growing interest in applying LLMs to automate and enhance penetration testing; however, they also reveal common challenges and diverging strategies. Several studies, such as those by Deng et al. [126,130] and Bianou et al. [133], proposed modular or multi-agent frameworks, demonstrating improved task decomposition and alignment with frameworks like MITRE ATT&CK. However, these systems face difficulties in handling complex scenarios or adapting to novel threats due to their reliance on predefined patterns or limited context retention. Other studies, such as those by Temara [130] and Fang et al. [129], examined specific capabilities of LLMs, such as reconnaissance or exploiting known vulnerabilities, but their findings highlight a dependency on structured input and the need for better validation mechanisms. Studies by Happe and Cito [132] and Happe et al. [41] also explored the use of LLMs as supportive tools for human testers; however, they emphasized concerns around ethical deployment and the models' inability to manage errors and maintain focus during tasks. Isozaki et al. [40] also proposed benchmarks to evaluate LLM performance, but their findings confirm that human oversight remains crucial for end-to-end testing. Generally, while these approaches show promise, limitations persist in scalability, reliability, and context management across tasks.

4.6. Digital Forensics

LLMs have shown significant promise in enhancing the efficiency and accuracy of digital forensics by automating complex tasks such as evidence search, anomaly detection, and report generation. Many studies have investigated their effectiveness in addressing traditional challenges in forensic investigations, with promising results in areas like ransomware triage, synthetic dataset generation, and cyber incident analysis. Yin et al. [134] provided an overview of how LLMs are transforming digital forensics, emphasizing their capabilities and limitations while highlighting the need for forensic experts to understand LLMs to maximize their potential in various forensic tasks. Similarly, Scanlon et al. [1] evaluated ChatGPT's impact on digital forensics, identifying low-risk applications, like evidence searching and anomaly detection, but also noting challenges related to data privacy and accuracy. Wickramasekara et al. [135] also explored the integration of LLMs into forensic investigations, highlighting their potential to improve efficiency and overcome technical and judicial barriers, though emphasizing the need for appropriate constraints to fully realize their benefits.

Bin Oh et al. [136] proposed volGPT, an LLM-based approach designed to enhance memory forensics, specifically in triaging ransomware processes. Their research demonstrates high accuracy and efficiency in identifying ransomware-related processes from memory dumps, providing more detailed explanations during triage than traditional methods. Similarly, Voigt et al. [137] combined LLMs with GUI automation tools to generate realistic background activity for synthetic forensic datasets. This innovation addresses the challenge of creating realistic "wear-and-tear" artefacts in synthetic disk images, improving forensic education and training. In addition, Michelet and Breiteringer [138] explore the use of local LLMs to assist in writing digital forensic reports, demonstrating that while LLMs

can aid in generating content, human oversight remains essential for accuracy and completeness, highlighting their limitations in complex tasks like proofreading. Bhandarkar et al. [139] evaluated the vulnerabilities of traditional DFIR pipelines in addressing the threats posed by Neural Text Generators (NTGs) like LLMs. Their study introduced a co-authorship text attack, CS-ACT, revealing significant weaknesses in current methodologies and emphasizing the need for advanced strategies in source attribution. Loumachi et al. [140] also proposed GenDFIR, a framework that integrates LLMs with RAG for enhancing cyber incident timeline analysis. By structuring incident data into a knowledge base and enabling semantic enrichment through LLMs, GenDFIR automates timeline analysis and improves threat detection.

Similarly, Zhou et al. [141] presented an LLM-driven approach that constructs Forensic Intelligence Graphs (FIGs) from mobile device data, achieving over 90% coverage in identifying evidence entities and their relationships. Also, Kim et al. [142] evaluated the application of LLMs, like GPT-4o, Gemini 1.5, and Claude 3.5, in analysing mobile messenger communications, demonstrating improved precision and recall in interpreting ambiguous language within real crime scene data. Sharma [143] proposed ForensicLLM, a locally fine-tuned model based on LLaMA-3.1-8B, which outperformed its base model in forensic Q&A tasks and showed high accuracy in source attribution. Xu et al. [144] conducted a tutorial exploring the potential of LLMs in automating digital investigations, emphasizing their role in evidence analysis and knowledge graph reconstruction. On the other hand, Cho et al. [145] discussed the broader implications of LLM integration in digital forensics, emphasizing the need for transparency, accountability, and standardization to fully realize their benefits. Table 7 summarizes the contributions and limitations of recent LLM-based systems in digital forensics in ascending order per year of publication.

Table 7. Summary of the contribution and limitations of recent LLM-based systems in digital forensics.

Citation	Year	Summary of Contributions	Limitations
Scanlon et al. [1]	2023	Evaluated the impact of ChatGPT on digital forensics, presenting strengths and limitations across various forensic tasks.	Requires expertise and raises privacy concerns.
Bin Oh et al. [136]	2024	Proposed volGPT, an LLM-based approach for triaging ransomware within memory forensics.	Effectiveness is limited to ransomware detection.
Voigt et al. [137]	2024	Proposed Re-imagen, a framework that uses LLMs to generate realistic background activity for synthetic forensic datasets.	Unable to generate realistic background activity in synthetic datasets.
Michelet and Breitnger [138]	2024	Investigated the use of local LLMs in assisting the writing of digital forensic reports.	Requires extensive human proofreading and validation.
Bhandarkar et al. [139]	2024	Evaluated the readiness of DFIR pipelines to handle threats posed by NTGs in the LLM era.	DFIR methods need more advanced defensive strategies.
Loumachi et al. [140]	2024	Proposed GenDFIR, a framework that utilizes LLMs and RAG to enhance cyber incident timeline analysis.	Lacks scalability and transparency analysis.

Table 7. Cont.

Citation	Year	Summary of Contributions	Limitations
Zhou et al. [141]	2024	Proposed an LLM-driven method to construct FIGs from mobile data, achieving high coverage in identifying evidence entities and relationships.	Requires further validation across diverse datasets.
Xu et al. [144]	2024	Explored the potential of LLMs in automating digital investigations through tutorials and case studies.	Complex to implement in real-world dynamic investigations.
Cho et al. [145]	2024	Discussed the integration of LLMs in DF, focusing on the importance of transparency and standardization.	Needs bias and interpretability fixes for deployment.
Yin et al. [134]	2025	Provided a comprehensive overview of LLMs in digital forensics, highlighting capabilities and limitations.	Needs expert understanding of LLM strengths and limits.
Wicramaskara et al. [135]	2025	Explored how integrating LLMs into DF investigations can address challenges and improve efficiency.	Need a controlled LLM integration in forensics.
Kim et al. [142]	2025	Evaluated LLMs in analysing mobile messenger communications, improving precision and recall in interpreting ambiguous language.	Dependent on the quality of prompt engineering and model selection.
Sharma et al. [143]	2025	Proposed ForensicLLM, a locally fine-tuned model enhancing forensic Q&A tasks and source accuracy.	Constrained by training data and computing power.

The reviewed studies demonstrate increasing interest in applying LLMs to digital forensics while also exposing key limitations. Several studies focus on task-specific applications, such as ransomware triage [136], mobile message analysis [142], and forensic report writing [138]. Other studies, such as those by Scanlon et al. [1] and Yin et al. [134], take a broader view, evaluating the overall impact and limitations of LLMs across digital forensic tasks. Common themes include the need for human oversight due to LLM hallucinations and errors [1,138], limited generalizability to diverse datasets [141], and challenges in prompt engineering and model selection [142]. While tools like GenDFIR [140] and ForensicLLM [143] show promise in enhancing timeline analysis and Q&A tasks, respectively, their effectiveness is constrained by scalability and training data limitations. Furthermore, the lack of standardization, transparency, and ethical controls is a recurring concern [135,145]. Generally, while LLMs offer promising potential in digital forensics, robust validation, domain adaptation, and expert supervision remain essential for reliable deployment.

Table 8 offers a comprehensive overview of popular LLMs and their applicability across the six cyber security domains that this paper is focusing on. It highlights the versatility and evolution of these models over time, showcasing their strengths and limitations in specific areas. While many models demonstrate utility in tasks like vulnerability detection and threat intelligence, fewer are readily equipped for niche applications, such as blockchain security and digital forensics. This variation reflects the need for task-specific fine-tuning and the integration of domain-relevant datasets to fully unlock each model's potential. All the models listed in the table can be applied; however, their effectiveness in these specialized areas is generally limited without fine-tuning. These models are not explicitly designed for such tasks, so they require adaptation using domain-specific datasets and objectives. With proper fine-tuning and contextual alignment, they can support targeted

functions, such as anomaly detection, smart contract analysis, threat pattern recognition, and forensic data interpretation, though their performance may still vary depending on the complexity of the task.

Table 8. Popular LLM models across six cyber security domains.

Model	Release Time	Open Source	Developer	LLMs in Cyber Security					
				VD	AD	CTI	BC	Pentest	DF
BERT	2018.1	Yes	Google	✓		✓	x	x	✓
GPT-2	2019.02	Yes	OpenAI	x	✓	✓	x	x	✓
RoBERTa	2019.07	Yes	Facebook AI	x	✓	✓	x	x	✓
T5	2019.1	Yes	Google	✓	✓	✓	x	x	x
DistilBERT	2019.1	Yes	Hugging Face	✓	✓	✓	x	x	x
CodeBERT	2020.02	Yes	Microsoft	✓	x	x	✓	✓	x
GPT-3	2020.04	No	OpenAI	x	✓	✓	x	x	✓
DeBERTa	2020.06	Yes	Microsoft	✓	✓	x	x	x	✓
GraphCodeBERT	2020.09	Yes	Microsoft	✓	x	x	✓	✓	x
CodeT5	2021.09	Yes	Google	✓	x	x	✓	✓	x
CodeGen	2022.03	Yes	Salesforce	✓	x	x	x	✓	x
UniXcoder	2022.03	Yes	Microsoft	✓	x	x	x	✓	x
InCoder	2022.04	Yes	Cohere	✓	x	x	x	✓	x
Flan-T5	2022.1	Yes	Google	✓	✓	✓	x	x	✓
LLaMa	2023.02	Yes	Meta	x	✓	✓	x		x
StarCoder	2023.05	Yes	Hugging Face	✓	x	x	x	✓	x
Falcon	2023.06	Yes	Hugging Face	✓	✓	✓	x	✓	x
CodeLLaMa	2023.08	Yes	Meta	✓			✓	✓	x
Mistral 7B	2024.01	Yes	Mistral AI	x	✓	✓	x	x	✓
Jamba	2024.03	Yes	Jamboreebrain	x	✓	✓	x	x	x
DBRX	2024.03	Yes	DeepBrain	✓	x	x	x	✓	x
MAP-Neo	2024.05	Yes	MAP AI	x	✓	✓	x	x	x
LLaMA 3.1	2024.07	Yes	Meta	✓	✓	✓	x	x	x
Minerva 7B	2024.11	Yes	Minerva AI	✓	✓	✓	x	x	x
BitNet b1.58 2B4T	2025.04	Yes	BitTech	✓	x	x	✓	✓	x

Across the six surveyed domains mentioned earlier, LLMs have introduced capabilities that extend well beyond the scope of traditional rule-based or statistical approaches. Numerous empirical studies show measurable gains in key performance metrics. For instance, LProtector [52] achieved State-of-the-Art F1 scores on C/C++ vulnerability detection, and AnomalyGPT [79] demonstrated superior detection of industrial defects compared to classical thresholds. Similarly, Nahmias et al. [38] reported that contextual vectors derived from LLMs outperformed traditional classifiers in spear-phishing detection. These results suggest that when applied with targeted domain knowledge, LLMs can meaningfully improve detection accuracy, reduce false positives, and deliver more context-aware security insights.

However, these advantages come with critical limitations. A consistent challenge across domains is the limited robustness of LLMs under adversarial, ambiguous, or high-stakes operational conditions. Studies, such as the one by Sovrano et al. [39], highlight position-based bias in LLM outputs, while Happe et al. [132] showed that GPT-4-turbo

struggles with privilege escalation tasks requiring multi-step reasoning. In CTI, tools like LMCloudHunter [95] and 0-CTI [103] improve automation but rely heavily on high-quality retrieval sources or STIX formats, which can break in noisy environments. Meanwhile, systems like PentestGPT [128,131] perform well in isolated tasks but still require human oversight for complex logic flows. These empirical findings highlight that while LLMs bring substantial innovations, they are not universally applicable nor fully reliable as stand-alone systems. Critical deployment risks, such as hallucination, cost, latency, and explainability, persist across all domains. Therefore, despite their promising trajectory, LLMs must be deployed within hybrid architectures that retain deterministic safeguards and enable human-in-the-loop oversight.

5. Opportunities for LLMs in Cyber Security

In addition to the opportunities for LLMs in cyber security discussed in the previous section by reviewing State-of-the-Art studies regarding vulnerability detection, anomaly detection, cyber threat intelligence, blockchain, penetration testing, and digital forensics, there are several emerging areas in cyber security where LLMs can play a pivotal role. This section explores these new opportunities, demonstrating the potential of LLMs in the evolving cyber security landscape.

5.1. Malware Analysis and Detection

Malware detection is vital in cyber security. Unlike traditional ML models that rely on known signatures, LLMs analyse context, behaviour, and patterns in large datasets. They can detect new or unknown malware variants using textual descriptions, code snippets, and network traffic, offering a more adaptive and intelligent detection approach [87]. LLMs such as GPT-3, BERT, and CodeBERT have demonstrated their usefulness in analysing code snippets and software behaviours to detect malware signatures and anomalies [91]. Their NLP capabilities allow for a contextual understanding of code and file behaviour, offering improved detection over traditional signature-based approaches. Rahali and Akhloufi [146] proposed MalBERTv2, a code-aware BERT-based model for malware detection, and demonstrated that it significantly outperformed traditional models by incorporating semantic features of code into the classification process.

LLMs are particularly effective when applied to dynamic analysis, where they can monitor and interpret the behaviour of a program during execution. Rondanini et al. [147] demonstrated that LLMs, when trained on large datasets of both benign and malicious behaviour, can predict new malware behaviours more accurately than traditional signature-based systems. LLMs' contextual analysis allows them to recognize subtle indicators of suspicious activity that may not align with conventional signatures but still indicate malicious intent. Furthermore, LLMs can generate malware code and simulate attack behaviours for cyber security training purposes. Liu et al. [70] demonstrated how LLMs were used to automatically generate sophisticated malware code for penetration testing, enabling security teams to test their defences against realistic and evolving threats.

5.2. Real-Time Phishing Detection

Phishing remains a major threat, often evading traditional filters. LLMs enhance detection by analysing email content and identifying suspicious elements, like misleading URLs, attachments, and writing styles, making them effective against evolving and sophisticated phishing tactics [32,73]. They can understand subtle cues, contextual anomalies, and linguistic patterns that suggest malicious intent. Jamal and Wimmer [73] demonstrated how fine-tuned LLMs significantly outperform traditional ML models in identifying phishing content, especially zero-day phishing emails, due to their contextual reasoning capabilities.

By continuously learning from new data and adapting to evolving phishing tactics, LLMs can provide an additional layer of protection against email-based threats. This proactive, context-driven approach is crucial in defending against the increasing sophistication of phishing schemes. Furthermore, studies have shown that LLMs can analyse entire email threads rather than single messages in isolation, allowing them to detect more nuanced social engineering attacks that unfold across multiple messages. For example, Nahmias et al. [38] utilized a fine-tuned LLM model to classify phishing emails with a focus on conversational flow and semantic coherence, achieving significant gains over static filters. Additionally, when integrated with SIEM systems, LLMs can help in real-time threat prioritization by flagging high-risk emails for immediate response, thereby reducing the burden on human analysts [7].

5.3. Automated Vulnerability Management

Automating vulnerability management with LLMs offers great potential. This process, which typically demands manual effort and expertise, can be streamlined by LLMs that analyse and remediate vulnerabilities. Trained on extensive datasets, LLMs can predict flaws in new code or systems, easing the burden on security teams and enhancing efficiency [48]. Wang et al. [58] showed that LLMs could analyse code for patterns that correspond to common vulnerabilities, such as SQL injection, cross-site scripting (XSS), or buffer overflows. The model's ability to recognize subtle coding practices that may lead to security holes helps identify vulnerabilities early in the development lifecycle.

LLMs are increasingly used to improve software security through secure code generation and automated vulnerability remediation. Pearce et al. [148] showed that Codex could detect vulnerable code patterns and suggest secure alternatives with greater accuracy than traditional static analysis tools. Ghosh et al. [67] demonstrated LLMs' ability to repair vulnerabilities, like buffer overflows, while explaining their fixes. Lin and Mohaisen [65] found that LLMs can recommend secure coding alternatives to mitigate risks, aiding developers without deep security expertise. Additionally, LLMs trained on large codebases can generate security-hardened code snippets, reducing human error and promoting secure development practices [54]. These capabilities highlight LLMs' potential to proactively support developers in writing and maintaining secure software.

5.4. Threat Intelligence Automation

LLMs can significantly enhance threat intelligence by automating the extraction of valuable insights from unstructured data sources, like blogs, news articles, and dark web forums, streamlining the process of identifying emerging threats [101]. A key application of LLMs is the analysis of OSINT to detect emerging threats. Shafee et al. [5] demonstrated how LLMs can process massive volumes of OSINT data to identify trends in attack methods, tools, and targets. By analysing social media posts, forums, and even code repositories, LLMs can highlight potential vulnerabilities or zero-day exploits before they become widely known. This early detection allows organizations to take preventive actions to mitigate these risks.

LLMs can also automate the creation of threat intelligence reports by summarizing key findings from raw data. Gao et al. [102] demonstrated that LLMs can take raw, unstructured threat data and generate structured reports, making it easier for cyber security professionals to digest and act on relevant intelligence. This capability is especially valuable in high-volume environments where security teams need to process large amounts of data quickly. In addition, fine-tuned LLMs can be used to classify and prioritize threats based on contextual relevance, helping analysts distinguish between benign anomalies and critical attack indicators [108]. For example, the use of transformer-based models has enabled

cyber security platforms to automatically cluster similar threat indicators and associate them with specific adversary tactics and techniques [95,108].

5.5. Social Media and Dark Web Monitoring

Cybercriminals frequently use social media and dark web forums to communicate, share tools, and coordinate attacks. LLMs can monitor these platforms to detect potential threats by identifying discussions about exploits, attack vectors, and vulnerabilities [12]. They can be trained to recognize illicit cyber activities, such as the sale of exploit kits or ransomware-as-a-service. Shah and Parast [110] demonstrated how LLMs can extract relevant intelligence from unstructured dark web data, aiding cyber security teams and law enforcement in identifying emerging threats. Additionally, LLMs can analyse social media posts to detect early signs of attacks, such as leaked data or targeted threats [149].

Beyond passive monitoring, LLMs can actively support cyber threat-hunting workflows by identifying disinformation campaigns, social engineering tactics, and recruitment efforts by malicious actors. These models detect linguistic markers and behavioural patterns linked to bots, troll farms, or coordinated inauthentic behaviour [95]. Fang et al. [129] used a fine-tuned GPT-based model to attribute social media content to known threat actors through linguistic fingerprinting and thematic clustering, which is particularly valuable for tracking APTs and state-sponsored campaigns. On the dark web, LLMs also help uncover relationships between threat actors by analysing language patterns, aliases, and interaction threads, revealing hidden networks and collaboration [124]. Social media and dark web monitoring is a promising area for LLMs in cyber security, offering opportunities to further develop automated threat detection by extracting actionable intelligence from unstructured and covert online communications.

5.6. Incident Response and Playbook Generation

LLMs can transform incident response by automating playbook creation and providing real-time, tailored recommendations. Kim et al. [142] demonstrated that LLMs can generate customized playbooks based on specific attack scenarios, enabling rapid, consistent responses and reducing human error. LLMs also support decision-making and coordination across teams and tools by analysing real-time data [140]. Additionally, Scanlon et al. [1] showed that LLMs can integrate with SOAR systems to automate mitigation actions, such as blocking malicious IPs or isolating affected systems, streamlining response efforts and improving efficiency.

Moreover, Wudaliet al. [150] demonstrated how LLMs could convert NIST or MITRE ATT&CK documentation into automated workflows for policy enforcement or compliance reporting. These models help bridge the gap between human-readable policy language and machine-executable logic. By translating regulatory and procedural documents into actionable formats, LLMs ensure the consistent application of security policies across systems. This is particularly valuable for organizations navigating complex compliance environments, where aligning technical controls with legal and industry standards is critical [94]. LLMs can also assist in continuously monitoring for policy violations and recommending remediation steps when deviations are detected [139]. Incident response and playbook generation represent a key opportunity for LLMs in cyber security, where further exploration could enhance automated, context-aware decision-making and streamline coordinated responses to evolving threats.

5.7. Advanced Penetration Testing and Red Teaming

Penetration testing and red teaming are crucial activities for identifying vulnerabilities and testing an organization's security defences. LLMs can play a role in automating these activities by generating attack simulations and analysing the effectiveness of

defences [132]. LLMs can simulate sophisticated attack strategies, including advanced persistent threats (APTs), which can test an organization's ability to withstand complex cyberattacks. Temara [130] showed that LLMs could generate detailed penetration testing scripts and attack scenarios based on the latest tactics, techniques, and procedures used by cybercriminals. In red-teaming exercises, LLMs can act as virtual adversaries, simulating attacks to identify gaps in security defences. As reported by Brown et al. [19], LLMs can generate realistic adversarial strategies, from phishing to exploiting vulnerabilities in web applications, providing security teams with a more comprehensive understanding of their weaknesses.

Moreover, LLMs are being used to augment automated reconnaissance and payload crafting, streamlining the preparation phases of red teaming. Happe et al. [41] demonstrated that fine-tuned LLMs can craft custom payloads, obfuscate malicious commands, and adapt exploits based on target system configurations. These capabilities enable red teams to execute context-aware attacks that mimic real-world adversaries, increasing the realism and impact of penetration tests. Additionally, in continuous security validation scenarios, LLMs can generate test cases dynamically to evaluate endpoint and perimeter defences against evolving threats. This shift toward LLM-driven red teaming transforms the landscape of proactive defence by enabling scalable, intelligent, and adaptive adversarial simulation that would otherwise require extensive manual effort and expertise [126,130].

In summary, LLMs are transforming cyber security by offering intelligent, context-aware solutions across several critical domains. They enhance malware analysis by interpreting code behaviour, identifying unknown threats, and simulating malware for testing. LLMs also outperform traditional filters in phishing detection through their deep linguistic understanding and ability to analyse entire email threads. For vulnerability management, LLMs predict and remediate flaws, suggest secure coding practices, and generate secure code. They automate threat intelligence by extracting actionable insights from unstructured sources like OSINT and dark web forums. LLMs also monitor social media and dark web communications, uncovering disinformation campaigns and threat actor networks. In incident response, they generate adaptive playbooks, translate policy documents into workflows, and integrate with SOAR systems for real-time mitigation. Furthermore, LLMs bolster penetration testing and red teaming by simulating complex attacks and crafting adaptive payloads. These diverse applications highlight the vast and evolving potential of LLMs to automate, scale, and strengthen cyber defence capabilities.

6. Challenges of LLMs in Cyber Security

Despite the promising potential of LLMs in cyber security, they also present significant challenges. This section explores these challenges and possible solutions that should be implemented for the successful integration of LLMs into cyber security practices.

6.1. Model Reliability and Accuracy

One of the most persistent concerns in cyber security applications of LLMs is model reliability. LLMs often suffer from "hallucinations", where they generate plausible but factually incorrect information. This is especially critical in security contexts, where accuracy can mean the difference between a successful defence and a breach [83]. According to Yan et al. [33], hallucinations in cyber security question-answering systems can mislead learners and practitioners alike, making LLM outputs unreliable without external validation. Additionally, due to the static nature of their training data, many LLMs lack up-to-date knowledge of emerging threats, tactics, and vulnerabilities. This makes them ill-suited for real-time threat intelligence unless augmented with retrieval systems or live data [113].

Furthermore, model reliability varies based on prompt structure, context, and domain-specific knowledge, making reproducibility a challenge. In high-stakes environments such as SOCs, where precision is critical, even minor inaccuracies in model output can mislead analysts or delay incident response [84]. To mitigate these issues, researchers have explored RAG approaches, where models pull in external, validated documents during inference [103,151]. Rajapaksha et al. [109] proposed CyberRAG, which combines LLM capabilities with domain-specific knowledge graphs to validate outputs, improving answer reliability. However, such methods still require continuous updates to knowledge bases and ontologies to remain effective in rapidly evolving threat landscapes.

6.2. Explainability and Transparency

Another major challenge is the lack of explainability and transparency in LLM behaviour. These models operate as “black boxes”, producing outputs without clearly indicating how they arrived at those decisions. In mission-critical applications like intrusion detection or automated incident response, this opacity can undermine trust and accountability [12]. Users and auditors may be unable to verify or understand why a particular alert was generated or why a certain response action was recommended. Zhou et al. [44] argue that the inability to trace model reasoning poses ethical and operational risks, particularly in regulated sectors. In cyber security, where evidence-based investigation is essential, the lack of transparency limits the usefulness of LLM-generated intelligence for forensic or legal purposes. Similarly, Hossain et al. [120] emphasized that the opaque nature of LLM decisions makes it difficult for security teams to trust and verify model-generated insights, particularly in environments where transparency is mandated, such as government or financial sectors [51].

Efforts to increase interpretability, such as attention heatmaps or prompt engineering, are not always effective or practical in complex cyber scenarios. For instance, the AISeckG ontology described by Agrawal et al. [9] helps contextualize and validate model responses, offering a path toward greater transparency. However, Explainable AI (XAI) techniques are still maturing and integrating them into LLMs at scale without compromising performance is a significant hurdle [84]. Until these models can provide interpretable reasoning behind their outputs, their integration into decision-making processes in cyber security will remain constrained.

6.3. Adversarial Use of LLMs

LLMs present a double-edged sword in cyber security, as their capabilities can also be weaponized by malicious actors. One growing concern is the generation of sophisticated malware through prompt engineering [151]. As demonstrated by Yan et al. [33], LLMs can be manipulated to write polymorphic code that mimics legitimate software and executes malicious payloads. Although providers have added safety filters, jailbreaks and prompt injections can often bypass these constraints. This makes LLMs a potential tool for low-skill attackers to develop functional malware with minimal effort [10]. Brown et al. [19] demonstrated how GPT-4 could be prompted to generate obfuscated code resembling known malware, bypassing basic detection systems. This raises alarm over LLMs being weaponized by threat actors for automated exploit development.

Another alarming application is in social engineering, where LLMs can generate highly convincing phishing emails or social media messages. The natural language capabilities of LLMs make them ideal tools for crafting convincing phishing emails, as demonstrated in a study by Meguro and Chong [89], where LLMs outperformed humans in generating deceptive content. Threat actors can use these models to craft personalized lures at scale, mimicking the style and tone of legitimate communications [72,73]. Prompt injection attacks

can also compromise systems that integrate LLMs into workflows, enabling attackers to manipulate output or extract sensitive data. These risks highlight the need for robust input sanitization and access controls, which are often lacking in early-stage LLM deployments.

6.4. Security and Privacy Risks

LLMs are prone to prompt leakage, where sensitive data embedded in queries can be inadvertently exposed in outputs or logs. This risk is particularly severe in settings involving private threat intelligence or incident response. Moreover, the training data of many LLMs may contain sensitive or proprietary information scraped from online sources, leading to unintentional data disclosure [35]. Prompt leakage, where sensitive inputs can be inferred or exposed via crafted queries, is a notable risk. Yan et al. [33] showed that LLMs can unintentionally memorize and regurgitate snippets of sensitive training data, including passwords or confidential information, especially if exposed during pretraining. This raises compliance issues under laws like GDPR or CCPA, particularly if user data is included without consent [13].

Data poisoning is another critical issue. Attackers can manipulate the training data or fine-tuning process to embed backdoors or biases into the model. Ferrag et al. [151] indicated that backdoored LLMs can produce specific outputs when triggered, allowing adversaries to covertly influence model behaviour. In cyber security, this could result in a model that ignores specific IoCs or misclassifies malicious code. Since LLMs rely heavily on large-scale, often uncrated data, ensuring dataset integrity is a monumental task. Techniques such as differential privacy, data auditing, and secure fine-tuning are being explored to mitigate these risks, but they require significant resources and expertise to implement effectively [11,122].

6.5. Ethics and Governance

Ethical considerations and governance frameworks are also essential when deploying LLMs in cyber security. A primary issue is the dual-use nature of these models; while they can be used defensively, they can also empower threat actors. This duality complicates discussions around open access, model release strategies, and responsible disclosure [57]. Kambourakis et al. [2] argue that governance must evolve to address the societal risks posed by foundation models, especially in sensitive domains. In cyber security, this means implementing rigorous access controls, usage policies, and audit mechanisms to prevent misuse. Also, Yao et al. [11] argue that governance in cyber security applications must ensure that LLMs are used responsibly, preventing misuse by users.

Moreover, regulatory compliance is an evolving landscape. LLMs must align with data protection regulations like GDPR, which mandates transparency, accountability, and user rights. However, enforcing these principles is challenging given the opaque nature of LLMs and the scale at which they operate [10]. For example, Article 22 of the GDPR grants individuals the right not to be subject to automated decisions with legal consequences, which may apply to automated LLM-based threat assessments [152]. Meeting such requirements demands significant investments in compliance tooling, documentation, and human oversight, all of which add to operational complexity.

6.6. Evaluation and Benchmarking

LLM evaluation in cyber security is limited by benchmarks that fail to capture complex, evolving threats, like polymorphic malware or obfuscated code, causing general-purpose models to underperform in domain-specific tasks [123]. Static datasets also struggle to reflect rapidly changing attack patterns. While CyberRAG uses ontology-based validation to align with cyber security rules [105], it still depends on human oversight for complex cases, highlighting the need for dynamic, domain-specific evaluation frameworks to en-

sure reliable and automated model assessment. Another critical issue lies in the lack of standardized evaluation metrics tailored to cyber security applications. Many studies use accuracy or F1-scores without considering the cost of false positives or the severity of false negatives in threat detection scenarios [13]. Additionally, benchmarking LLMs across different subfields, such as vulnerability detection, threat intelligence summarization, and anomaly detection, requires nuanced, task-specific datasets that are rarely available due to sensitivity or proprietary concerns [123]. This scarcity of high-quality, annotated, and up-to-date data severely hampers reproducibility and cross-comparison, making it difficult to gauge true performance or progress in applying LLMs to cyber security.

6.7. Resource and Infrastructure Demands

LLMs are computationally intensive, requiring significant hardware for both training and inference. In cyber security environments where real-time processing is crucial, the latency introduced by large models can hinder operational effectiveness. Deploying LLMs for real-time threat detection may not be feasible in low-resource environments, like edge devices or air-gapped systems [12,88]. Moreover, maintaining an up-to-date, secure deployment pipeline for LLMs adds to the operational burden. This includes periodic retraining with fresh data, validating security updates, and ensuring compliance with internal and external standards. These challenges may limit LLM adoption in organizations with limited technical infrastructure or cyber security expertise [89].

Figure 5 shows the radar chart of the key challenges associated with the use of LLMs in cyber security, highlighting the severity of each issue on a scale from 0 to 10. The graph identifies seven major areas of concern that were discussed earlier. The most critical challenge is the adversarial use of LLMs, rated at the maximum score of 10, emphasizing the serious threat posed by malicious exploitation, such as the generation of malware, phishing content, and prompt injection attacks. Security and privacy risks (9) and model reliability and accuracy (9) also rank high, indicating substantial concerns regarding the trustworthy and safe deployment of these models, particularly in real-time or sensitive environments. Explainability and transparency (8) further reinforce this issue, as opaque model reasoning can exacerbate the risks of hallucinations and erode trust in automated decision-making. The challenge of evaluation and benchmarking received the lowest score (6), suggesting that while it remains important, it is currently perceived as less urgent compared to direct threats and vulnerabilities.

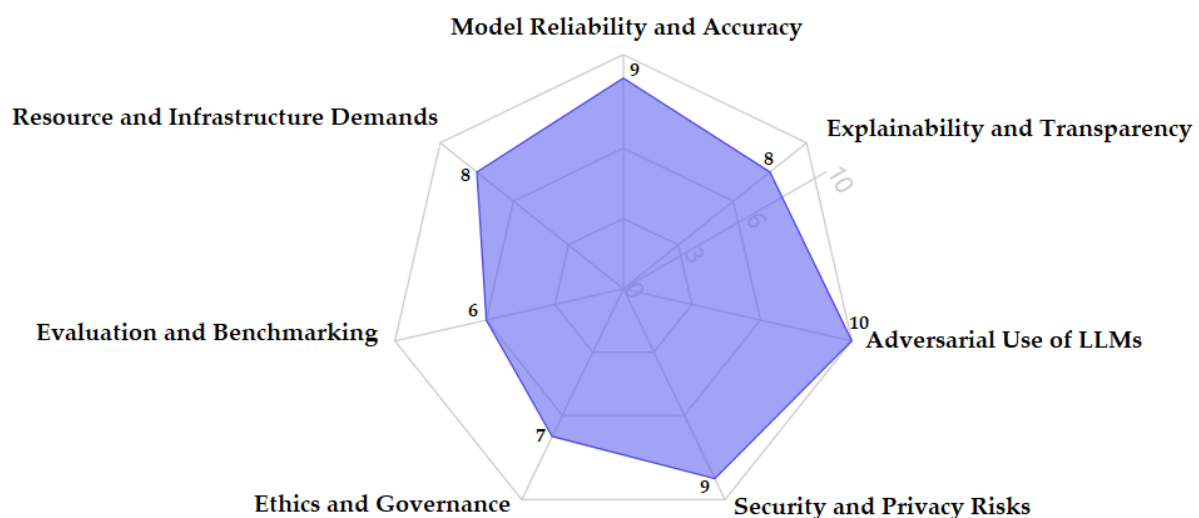


Figure 5. A radar chart of key challenges of LLMs in cyber security highlights the severity of each issue on a scale from 0 to 10.

Beyond simply ranking these challenges, the radar chart offers insight into their interdependence. For example, poor model reliability often stems from a lack of transparency, and both can contribute to heightened security risks. Similarly, adversarial misuse is closely tied to gaps in governance and the absence of robust input validation. The moderate rating for ethics and governance (7) reflects the growing recognition of dual-use concerns and regulatory implications but also highlights the need for more mature policy and oversight mechanisms. The relatively high score for resource and infrastructure demands (8) illustrates how technical feasibility, such as latency constraints and computational requirements, can significantly impact deployment in operational settings. Finally, although the challenge of evaluation and benchmarking received the lowest score, it plays a foundational role; without domain-specific benchmarks and robust metrics, it becomes difficult to systematically address other challenges.

7. LLMs in Cyber Security Education

The digital transformation driven by AI, cloud computing, and IoT has significantly broadened the cyber threat landscape, increasing the global demand for cyber security professionals. In the UK alone, there is a notable shortfall of approximately 11,200 cyber security specialists, with 37% of businesses lacking essential in-house skills [153]. This skills gap poses a serious threat to national infrastructure, economic resilience, and personal data privacy. To address this, UK higher education institutions have introduced a variety of cyber security degree programmes at both undergraduate and postgraduate levels. These are closely aligned with professional frameworks such as the Cyber Security Body of Knowledge (CyBOK) [154] and government-backed schemes like the NCSC-certified degree programme [155]. Such alignment ensures that graduates acquire both foundational theories and practical competencies needed in the industry. As cyber threats and technologies evolve, universities must adapt by integrating emerging innovations, like AI and LLMs, into their curricula while promoting inclusive and flexible learning opportunities. Bridging the skills gap is a multifaceted challenge requiring educational reform and societal engagement.

Cyber Security Education (CSE) faces a multitude of challenges, many of which are rooted in the rapidly evolving nature of the field, a persistent skills shortage, and the complexity of delivering hands-on, interdisciplinary learning [1,9,156,157]. These challenges can be summarized as follows:

- **Curriculum Lag:** Academic courses often fail to keep pace with fast-changing industry threats due to slow update and validation processes [9].
- **Lecturer Shortage:** There is a limited pool of educators with both academic and real-world cyber security expertise, especially in this specialized area [1].
- **Practical Learning Barriers:** Many institutions lack the infrastructure for hands-on training, such as cyber labs and ranges, which are expensive and complex to maintain [156].
- **Steep Learning Curve:** The technical breadth of cyber security can overwhelm students, particularly those from non-technical backgrounds [156].
- **Assessment Difficulties:** Ensuring fairness and academic integrity in practical assessments is challenging amid the rising use of online solutions [157].
- **Lack of Interdisciplinary Content:** Curricula often overemphasize technical skills, neglecting essential areas like policy, law, and ethics [156,157].
- **Access and Equity:** Financial, geographic, and institutional barriers prevent many students from accessing high-quality cyber security education [157].

7.1. Related Work of LLMs in CSE

LLMs are revolutionizing cyber security education by enabling interactive, scalable, and personalized learning experiences. As the cyber security landscape becomes increasingly complex and dynamic, LLMs offer promising solutions to bridge knowledge gaps, automate content creation, and enhance educational support. Several recent studies have explored their application in this domain. Zhao et al. [156] introduced CyberRAG, an ontology-aware RAG system that enhances question-answering by integrating domain-specific knowledge, thereby reducing hallucinations and improving response accuracy. Agrawal et al. [157] also proposed CyberQ, a system that combines LLMs with knowledge graphs to generate relevant questions and answers for cyber security education. This approach automates the creation of educational content, facilitating more dynamic and responsive learning environments. CyberQ complements CyberRAG by focusing on content generation, whereas CyberRAG emphasizes answer validation. Further extending this work, Zhao et al. [158] developed CyberBOT, a chatbot deployed in a graduate-level course that validates LLM outputs using course-specific materials and cyber security ontologies, demonstrating the practical value of structured reasoning in educational contexts.

Shepherd [159] examined a UK master's-level cyber security program, uncovering key vulnerabilities in assessments due to generative AI misuse, especially in projects and reports. Factors like block teaching and a largely international student body heightened these risks. The study advocates for LLM-resistant assessment designs, AI detection tools, and an ethics-focused academic culture. Also, Ohm et al. [160] conducted a three-year study on ChatGPT's impact across different student cohorts. While weekly assignment performance remained stable, final exam scores declined with increased ChatGPT usage. Despite a mandatory disclosure policy, 80% of students admitted anonymously to using ChatGPT; however, none reported it officially. This highlights growing concerns around honesty, dependency, and maintaining integrity in AI-supported education.

Nizon-Deladoueille et al. [161] evaluated six leading LLMs on penetration testing tasks, finding models like GPT-4o mini and WhiteRabbitNeo particularly effective in offering context-aware guidance. This highlights LLMs' capability to support hands-on technical training. Al-Dhamari and Clarke [162] also focused on enhancing cyber security awareness programs through GPT-driven personalization. Their work demonstrated that adaptive, profile-based training significantly improves learner engagement and outcomes, suggesting LLMs can address the limitations of static, one-size-fits-all instruction. Extending the application to support underrepresented learners, Wang et al. [163] proposed CyberMentor, a platform using agentic workflows and RAG to offer tailored mentorship and skills guidance. Tann et al. [164] also evaluated leading LLMs, including ChatGPT, Bard, and Bing, in professional certification exams and Capture-The-Flag (CTF) challenges. They found ChatGPT effective in answering factual multiple-choice questions (82% accuracy) but limited in conceptual reasoning. Performance varied across different CTF challenges, with prompt engineering playing a significant role. The study also highlighted vulnerabilities in LLM safety mechanisms, revealing how jailbreak prompts can bypass ethical safeguards.

Shao et al. [165] explored the effectiveness of LLMs in solving CTF challenges using both human-in-the-loop (HITL) and fully automated workflows. Their results show that LLMs outperform average human participants, highlighting their potential for automated cyber security training and assessment. This study highlights the ability of LLMs to solve real-world security puzzles, paving the way for their use in cyber security education. In addition, Nelson et al. [166] proposed SENSAT, an AI-powered tutoring system deployed in an applied cyber security curriculum. SENSAT uses LLMs to deliver real-time, context-aware feedback by integrating with students' terminals and files, significantly improving problem-solving efficiency across over 15,000 sessions. Its scalability, comparable in cost to

a single teaching assistant, shows promise for broad educational reach. Additionally, Yamin et al. [8] proposed CyExec, a framework that uses GPT models and RAG to dynamically craft complex, adaptive cyber security scenarios. By turning LLM hallucinations into creative leverage, CyExec offers diverse and realistic threat simulations rigorously validated through expert review. Park and Simmons [167] also explored the use of generative AI in cyber security education to address challenges like outdated materials, diverse student needs, and the absence of teaching assistants. It highlights AI's potential to create dynamic, personalized learning environments but also raises concerns about privacy, data ownership, and transparency in its implementation. Table 9 summarizes the contributions and limitations of recent LLM-based systems in cyber security education in ascending order per year of publication.

Table 9. Summary of the contribution and limitations of recent LLM-based systems in cyber security education.

Citation	Year	Summary of Contributions	Limitations
Tann et al. [164]	2023	Evaluated LLMs in certification exams and CTF challenges, showing ChatGPT excels in factual questions but struggles with conceptual reasoning.	Performance varies across CTF challenges, and jailbreak prompts bypass ethical safeguards.
Zhao et al. [156]	2024	Proposed an ontology-aware CyberRAG system to improve QA in cyber security education by integrating domain-specific knowledge.	Relies heavily on the quality and completeness of predefined ontologies.
Agrawal et al. [157]	2024	Developed CyberQ, a system that leverages LLMs and knowledge graphs to automatically generate questions and answers to enhance learning dynamics.	Requires frequent updates to knowledge graphs to maintain question relevance.
Ohm et al. [160]	2024	Empirically evaluated the impact of ChatGPT usage, revealing unchanged weekly performance but deteriorating exam results and low disclosure compliance.	Results are context-specific, and long-term cognitive impact is not addressed.
Nizon-Deladoueille et al. [161]	2024	Evaluated 6 LLMs in 15 penetration testing tasks, identifying GPT-4o mini and WhiteRabbitNeo as top performers for educational support.	Broader applicability requires further research.
Al-Dhamari & Clarke [162]	2024	Proposed GPT-integrated CSAT programs to deliver personalized and dynamic cyber security training, enhancing engagement and effectiveness.	Long-term efficacy and potential AI biases need further investigation.
Shao et al. [165]	2024	Evaluated LLMs in solving CTF challenges with HITL and fully automated workflows, showing LLMs outperform humans in solving security puzzles.	The results are limited to a specific set of CTF challenges, so generalizability needs further exploration.
Yamin et al. [8]	2024	Proposed CyExec, utilizing LLMs and RAG to create dynamic, complex cyber security training scenarios.	Lack of realism and requires continuous validation.
Park and Simmons [167]	2024	Investigated LLMs in cyber security education for creating dynamic, personalized learning environments.	Concerns include privacy, data ownership, and transparency.

Table 9. Cont.

Citation	Year	Summary of Contributions	Limitations
Zhao et al. [158]	2025	Developed a chatbot in a graduate course that uses an RAG pipeline and course-specific ontology to validate LLM-generated responses.	Scalability to different courses and topics may be constrained by ontology and content curation.
Shepherd [159]	2025	Investigated LLM misuse in a UK cyber security master's program, highlighting vulnerabilities in assessments and proposing mitigation strategies.	Focused on a single institution, so a broader applicability needs further investigation.
Nelson et al. [166]	2025	Proposed SENSAT, an AI tutoring system using LLMs for scalable, personalized feedback in cyber security education, improving problem-solving efficiency.	Tested in one university, so further evaluation is needed across different contexts.
Wang et al. [163]	2025	Proposed CyberMentor, an AI-powered platform offering personalized support for non-traditional cyber students in learning and career development.	Requires broader validation and continuous content updates for sustained relevance.

The studies reviewed emphasize the expanding role of LLMs in cyber security education, particularly in enhancing engagement, personalizing learning, and automating content generation. A recurring similarity across various studies, such as those by Agrawal et al. [157], Yamin et al. [8], and Zhao et al. [156], is the integration of knowledge graphs or ontologies to improve question generation or response accuracy. However, this reliance introduces limitations around scalability and the need for regular content updates. Several studies, such as those by Tann et al. [164] and Shao et al. [165], highlight the effectiveness of LLMs in challenge-based scenarios, like CTFs, though they also caution against generalizing findings due to dataset or domain specificity. In addition, studies, like those by Al-Dhamari and Clarke [162], Nelson et al. [166], and Wang et al. [163], focus on LLM-driven training and mentoring platforms, showcasing the potential for scalable and inclusive learning while recognizing risks related to AI bias, privacy, and the need for long-term validation. Shepherd [159] and Ohm et al. [160] offer more critical views, raising concerns about assessment integrity and cognitive dependency.

7.2. Opportunities for LLMs in CSE

LLMs offer innovative opportunities in cyber security education by enabling personalized, interactive, and scalable learning. They can automate content creation, simulate real-world scenarios, and provide adaptive support. LLMs can offer several opportunities/benefits to cyber security education, including the following:

7.2.1. Personalized Learning and Tutoring

LLMs present a novel approach to personalized learning by acting as intelligent tutors capable of delivering adaptive, real-time feedback. Through context-aware assistance, these models can cater to diverse learner profiles, offering both foundational and advanced explanations tailored to individual needs [10]. Al-Dhamari and Clarke [162] demonstrated that GPT-driven personalization enhances cyber security awareness by increasing engagement through profile-based content. Similarly, Wang et al. [163] developed CyberMentor, a RAG-powered platform that provides tailored guidance to under-represented students, emphasising the role of agentic workflows in skill development.

7.2.2. Curriculum Enhancement & Content Generation

LLMs facilitate efficient curriculum development by automating the generation of cyber security content, such as case studies, lecture materials, and assessments [50]. Agrawal et al. [157] developed CyberQ, a system that utilizes LLMs with knowledge graphs to dynamically produce relevant questions and answers, thereby enriching learning environments with responsive and diverse content. Park and Simmons [167] emphasized how generative AI can address the limitations of static curricula, particularly the rapid obsolescence of teaching materials in cyber security, by enabling continuously updated and student-centred educational resources.

7.2.3. Practical Skills Development

Beyond theoretical learning, LLMs contribute meaningfully to hands-on skills development. Nizon-Deladoueille et al. [161] evaluated multiple LLMs and found models like GPT-4o mini to be particularly adept at offering context-sensitive support in penetration testing, helping students grasp complex command-line tasks. Shao et al. [165] also highlighted the effectiveness of LLMs in solving CTF challenges through both human-in-the-loop and automated approaches, demonstrating their potential to foster real-world cyber security competencies in training environments.

7.2.4. Assessment and Feedback Automation

LLMs can streamline assessment by generating individualized questions, automatically grading submissions, and offering rubric-aligned feedback. Zhao et al. [158] implemented CyberBOT in a graduate cyber security course, where it validated LLM-generated answers using domain-specific ontologies and course materials, ensuring accurate and contextually grounded feedback. However, concerns remain regarding LLM misuse in assessments. Shepherd [159] identified vulnerabilities in assessment integrity, especially in project-based tasks, and stressed the need for LLM-resilient assessment designs to maintain academic honesty and mitigate the risks posed by generative AI [50].

7.2.5. Promoting Cyber Awareness

Promoting foundational cyber security awareness among students is another area where LLMs excel. LLMs demonstrated that adaptive, LLM-driven training modules significantly improve awareness and retention compared to traditional methods [104]. These systems not only explain best practices but can also simulate cyber incidents, allowing learners to experience realistic threat scenarios [162]. Park and Simmons [167] advocate for LLMs as tools to address diverse learner needs, suggesting their application in broad-based cyber hygiene education across academic contexts. Feng et al. [64] also showed that integrating ChatGPT into cyber security MOOCs increased student engagement and conceptual understanding, particularly in explaining abstract concepts like access control or encryption protocols.

7.2.6. Research and Innovation

LLMs play a pivotal role in advancing research and innovation in cyber security education by facilitating the exploration of new threats, trends, and technologies. These models can process vast amounts of data, helping researchers analyse emerging security challenges and generate novel solutions [110]. LLMs also support innovation by enabling the rapid creation of educational content, generating research ideas, and automating repetitive tasks, thus accelerating the pace of academic progress. Additionally, LLMs enhance collaboration by fostering the sharing of insights and knowledge across research

teams, making them invaluable tools in the ongoing development of cutting-edge cyber security practices and solutions.

7.2.7. Peer Learning Support

LLMs empower educators as co-creators, instructional aides, and scalable mentors. Nelson et al. [166] presented SENSAL, an AI-powered tutoring system integrated into a cyber security curriculum, delivering real-time feedback during student labs. With over 15,000 interactive sessions and operational costs equivalent to a single teaching assistant, SENSAL illustrates the scalability and efficiency of LLM-supported faculty assistance. This reduces workload, supports inclusive teaching, and improves peer learning environments by supplementing instructional gaps, especially in large or diverse cohorts. Beyond automation, LLMs foster enriched peer learning environments by offering consistent support, addressing common misconceptions, and enabling timely interventions tailored to diverse learner needs.

7.3. Challenges of LLMs in CSE

Despite their potential, the use of LLMs in cyber security education comes with several challenges, which include the following:

7.3.1. Accuracy and Hallucination

One of the most critical challenges in using LLMs in cyber security education is their tendency to “hallucinate”, which produces confident-sounding but factually incorrect information. This issue becomes particularly problematic in a domain like cyber security, where precision and correctness are vital [81,107]. Students may rely on these outputs without validating the information, leading to the internalization of misconceptions or faulty practices. This not only undermines the quality of learning but can also be dangerous if applied in real-world security contexts. Tools like CyberRAG [156] have attempted to mitigate this by integrating domain ontologies for validation, but hallucinations remain a persistent issue, especially when LLMs are used without appropriate safeguards or human oversight.

7.3.2. Ethical and Academic Integrity

The integration of LLMs in educational contexts has raised significant ethical concerns, particularly regarding their misuse by students. Tools like ChatGPT can easily generate entire assignments, reports, or code snippets, enabling students to bypass critical learning processes. Research by Shepherd [159] and Ohm et al. [160] highlights that many students use LLMs without disclosure, which jeopardizes academic integrity and skews assessment outcomes. Furthermore, while some institutions have implemented disclosure policies, enforcement is inconsistent and often ineffective [111]. The challenge is cultivating a culture of responsible AI use that prioritizes transparency, ethics, and learning, alongside technical solutions, like AI detection tools and LLM-resistant assessments.

7.3.3. Assessment Validity

Traditional assessment formats, such as essays, written reports, and basic coding tasks, are increasingly vulnerable to automation by LLMs. Students can now generate high-quality responses with minimal effort, making it difficult for educators to distinguish between original work and AI-generated content. This erodes the reliability of assessment as a tool to evaluate true learning and skill development [102,164]. To address this, educators are being urged to adopt LLM-resilient assessment formats that emphasize critical thinking, problem-solving, and collaboration. These may include oral exams, live coding, group projects, and

scenario-based tasks. The key challenge is ensuring these redesigned assessments remain scalable, inclusive, and aligned with learning outcomes in diverse and often large cohorts.

7.3.4. Data Privacy and Security

Deploying LLMs in educational environments raises critical concerns around data privacy, security, and compliance. When students or educators input prompts that include sensitive personal information, institutional data, or even real cyber security incidents, this information could be processed and retained by third-party AI providers. This is particularly concerning in jurisdictions governed by strict data protection laws, such as the GDPR [152]. Institutions must therefore carefully evaluate the terms of use of commercial LLMs and consider privacy-preserving alternatives or on-premise deployments. Transparency about data handling, student consent, and secure integration with learning platforms is essential to prevent accidental exposure of confidential information and ensure compliance with ethical and legal standards.

7.3.5. Limited Conceptual Reasoning

Despite their impressive performance in producing syntactically correct and well-structured outputs, LLMs often lack deep conceptual reasoning abilities. They are primarily pattern recognition systems, trained on vast amounts of data without true understanding. Tann et al. [164] found that while models like ChatGPT performed well on factual multiple-choice questions, they struggled with tasks requiring deeper insight, contextual adaptation, or logical reasoning. This limitation becomes a barrier in advanced cyber security topics, where understanding abstract principles, system-level thinking, and contextual decision-making is essential [80]. Relying too heavily on LLMs for instruction or self-study in such areas may hinder the development of higher-order cognitive skills and critical thinking in students.

7.3.6. Instructor Readiness and Training

A significant barrier to effective LLM integration in cyber security education is the readiness of instructors to use these tools thoughtfully and critically. Many educators are still unfamiliar with prompt engineering, AI-assisted content validation, and the pedagogical implications of LLMs. Without proper training, instructors may either misuse the technology or avoid it altogether, leading to inconsistent learning experiences for students. Moreover, the fast-evolving nature of LLMs demands ongoing professional development to keep up with their capabilities, limitations, and ethical considerations. Institutions must, therefore, invest in capacity-building initiatives that equip faculties with the skills and confidence to harness LLMs in ways that enrich, rather than compromise, the learning environment.

8. Open Issues and Future Directions

The integration of LLMs into cyber security and cyber security education is promising but still emerging, with several open issues and future directions that researchers must explore. These include the following:

8.1. Trustworthiness and Hallucination Mitigation

A major concern with deploying LLMs in cyber security is their tendency to produce hallucinated or factually incorrect information. In high-stakes domains, such as digital forensics, malware analysis, or threat detection, even small inaccuracies can compromise security protocols. This issue becomes more pronounced when LLMs are used without domain-specific grounding, leading to responses that may appear confident but lack technical validity or relevance. While tools like CyberRAG [107,156] have attempted to mitigate

this through ontology-aware RAG, much work is still needed to enhance output reliability. Future research should focus on integrating structured knowledge sources into LLM pipelines, using adaptive validation for trustworthiness, and exploring hybrid models combining LLMs with symbolic reasoning to improve transparency and reduce errors, especially in cyber security education. Also, future research should involve user-centred evaluations and adversarial testing protocols to assess the reliability of hallucination mitigation techniques in real-world scenarios.

8.2. Privacy and Security

A critical issue in the deployment of LLMs across cyber security domains is the potential for unintended data leakage and exposure of sensitive information. These models, when integrated into threat analysis systems, incident response platforms, or automated security tools, often process logs, configurations, or code snippets that may contain proprietary or confidential data. The risk is particularly severe when using third-party LLMs via cloud-based APIs, where prompts and outputs could be cached or monitored, inadvertently exposing organizational vulnerabilities [77]. This challenge extends to training and simulation environments, including educational settings, where student data or simulated attack scenarios may also be compromised [164]. More research is needed to develop privacy-preserving LLM architecture for secure, local deployment, using techniques like federated learning, differential privacy, and zero-trust access controls. Additionally, AI governance frameworks must be created to ensure responsible LLM integration in both industry and academic cyber security settings. Future research should utilize empirical case studies and simulation-based testing with federated architectures to assess the effectiveness of privacy-preserving LLM deployments.

8.3. Explainability

A key limitation of current LLMs is their lack of explainability, which undermines their credibility in both teaching and professional cyber security workflows. Students and practitioners need more than just correct answers; they need to understand the reasoning behind those answers to develop critical thinking and investigative skills [159]. This black-box nature is especially problematic in scenarios like policy auditing, secure coding, or threat mitigation, where step-by-step justification is crucial [161]. Future research should prioritize the development of interpretable LLMs that can articulate their reasoning processes in human-readable formats. Techniques such as chain-of-thought prompting, attention heatmaps, or integration with argumentation frameworks could help users trace the model's logic. Embedding rationale generation directly into educational tools would empower learners to question, refine, and validate AI guidance, fostering deeper understanding and accountability. Also, user studies and think-aloud protocols can be used to evaluate how effectively LLMs convey their reasoning and enhance user understanding.

8.4. Adversarial Robustness

LLMs are vulnerable to adversarial inputs and jailbreak prompts that can manipulate their outputs or bypass ethical safeguards, posing a serious risk in both cyber security education and practice [165]. This problem is particularly critical in environments like CTF challenges or simulation-based training, where learners may experiment with edge-case queries or adversarial code [164]. Such vulnerabilities not only risk misinformation but can also normalize harmful behaviour if left unchecked. Future research should focus on developing prompt sanitization, adversarial training, and red-teaming frameworks for education while incorporating safety constraints and ethical reasoning in LLMs. Proactive monitoring using anomaly detection can prevent misuse and foster responsible AI engagement in instructional settings. Also, future research should employ red-teaming exercises

and controlled adversarial trials to evaluate model resilience and identify failure modes under attack conditions.

8.5. Standardization and Benchmarks

The lack of standardized evaluation benchmarks for LLMs in cyber security poses a barrier to rigorous comparison and progress tracking [57]. Without consistent metrics and datasets, it is difficult to assess which models perform best across tasks such as phishing detection, malware triage, or cyber threat intelligence extraction [13]. Researchers should focus on building community-driven benchmarks that reflect real-world complexity and adversarial conditions. This includes curating diverse and dynamic datasets, defining robust metrics for utility and robustness, and fostering open competitions to drive innovation. Community-driven design workshops and open benchmarking competitions could also facilitate the development of standardized, task-specific evaluation frameworks.

8.6. Regulatory Compliance

The deployment of LLMs in cyber security intersects with complex regulatory frameworks around data protection, intellectual property, and algorithmic accountability. Using LLMs for threat detection or compliance auditing may involve the processing of personally identifiable information (PII), proprietary threat signatures, or sensitive logs, raising serious concerns under laws like GDPR, CCPA, and NIS2 [40]. There is also growing pressure to ensure algorithmic transparency and explainability in automated decision-making systems. To navigate this evolving landscape, future research should integrate legal and ethical compliance into the LLM design lifecycle, focusing on data minimization, consent-aware data handling, and audit trails. Interdisciplinary design research and policy analysis are recommended to co-develop compliant AI systems with input from legal and cyber security experts. Collaboration between cyber security experts, legal scholars, and ethicists is essential for creating regulation-aligned AI tools and frameworks that support ethical innovation in LLM applications.

8.7. Standardized Curriculum Framework

Despite the promising use cases of LLMs in platforms like CyberQ [157] and SENSAI [166], there is currently no consensus on how best to integrate LLMs into cyber security curricula. Without standardized frameworks, their use remains fragmented and inconsistent, making it difficult to align AI-powered tools with established learning outcomes, accreditation standards, or ethical guidelines. This gap can lead to unequal learning experiences and complicate assessment design. To overcome this, future directions should focus on the co-design of curriculum-aligned LLM platforms in collaboration with educators, accrediting bodies, and industry partners. This includes developing instructional blueprints for scaffolded LLM use, ethical disclosure policies, and AI literacy modules that prepare students to work responsibly with generative tools. Creating a repository of best practices, assessment templates, and validated prompt libraries can further aid in seamless and equitable integration.

8.8. Augmented Intelligence

While LLMs offer the potential to improve the cyber security workforce shortage by automating routine tasks such as log analysis, report generation, and incident summarization, over-reliance on automation may contribute to skills degradation or foster a false sense of expertise among junior analysts [158]. This concern extends to education, where students might lean on generative models without fully developing critical thinking or hands-on technical skills. Future research should focus on augmented intelligence, designing LLMs as collaborative tools that enhance human judgment. In education, scaffolded

learning designs should integrate LLMs for feedback while preserving student input. In industry, upskilling programs and transparent AI policies are crucial to maintaining analyst competence alongside AI augmentation.

8.9. Evaluation Frameworks

As the adoption of LLMs in cyber security continues to expand, there is still a lack of standardized frameworks for evaluating the maturity, reliability, and pedagogical effectiveness of these models across diverse operational and educational settings [5]. Current studies often report outcomes in isolated contexts without consistent metrics for impact, scalability, or real-world feasibility [131,166]. Future research should focus on developing comprehensive evaluation frameworks that account for technical performance (e.g., accuracy, robustness, and interpretability), deployment readiness (e.g., integration complexity and resource requirements), and educational value (e.g., learning outcomes, engagement, and inclusivity). Such frameworks would guide researchers in benchmarking tools, assist practitioners in selecting appropriate solutions, and support educators in aligning LLM-based tools with pedagogical goals. Cross-institutional studies, expert validation, and stakeholder engagement will be essential in building robust and context-aware assessment models.

Table 10 summarizes the open issues and suggested future research directions. In summary, the integration of LLMs into cyber security and education presents significant promise but also introduces several critical open issues that require further research. The key challenges include ensuring trustworthiness and mitigating hallucinations, protecting sensitive data through privacy-preserving architectures, and addressing the lack of transparency in model outputs. Adversarial vulnerabilities and misuse highlight the need for robust safety measures and ethical frameworks. The absence of standardized benchmarks and evaluation protocols hinders the consistent assessment of model performance, while regulatory compliance adds complexity due to evolving data protection laws. Furthermore, the lack of a standardized curriculum framework limits the effective integration of LLMs into education, risking fragmented learning experiences. Concerns also persist around over-reliance on AI, which may diminish critical thinking and practical skills. To address these, future work should focus on building interpretable, secure, regulation-compliant LLM systems that support augmented intelligence and include robust evaluation and curricular integration frameworks.

Table 10. Summary of open issues and suggested future research directions for LLMs in cyber security.

Open Issue	Description of Research Gap	Suggested Research Direction
Trustworthiness and Hallucination Mitigation	LLMs may generate inaccurate or misleading content, especially without domain-specific grounding.	Integrate structured knowledge, adaptive validation, and symbolic reasoning; use user-centred and adversarial testing.
Privacy and Security	LLMs risk exposing sensitive data when processing logs or prompts via third-party APIs.	Develop federated, privacy-preserving deployments and simulate data governance in training environments.
Explainability	Lack of reasoning transparency undermines credibility in teaching and security workflows.	Design interpretable outputs and embed rationale generation; assess effectiveness through user studies.

Table 10. Cont.

Open Issue	Description of Research Gap	Suggested Research Direction
Adversarial Robustness	LLMs are susceptible to jailbreaks and adversarial prompts, especially in hands-on training.	Develop prompt sanitization, red-teaming, and anomaly detection frameworks tailored for instructional use.
Standardization and Benchmarks	Inconsistent metrics and datasets hinder fair evaluation of LLM capabilities.	Curate domain-specific benchmarks and host competitions for comparative testing.
Regulatory Compliance	Use of LLMs intersects with evolving legal and ethical standards (e.g., GDPR and CCPA).	Apply policy analysis and interdisciplinary design to align LLMs with compliance and auditability needs.
Standardized Curriculum Framework	No unified guidance exists for integrating LLMs into accredited cyber security education.	Co-develop blueprints, ethical policies, and prompt libraries with educators and accrediting bodies.
Augmented Intelligence	Automation risks weakening analysts' and students' practical skills and judgment.	Design LLMs as collaborative tools with scaffolded learning and transparent feedback for learners and analysts.
Evaluation Frameworks	No comprehensive models exist for assessing reliability, integration readiness, or pedagogical impact.	Build evaluation frameworks incorporating technical, deployment, and educational metrics validated cross-institutionally.

9. Conclusions

This paper presents a comprehensive review of the current and emerging applications of LLMs in cyber security, spanning both operational and educational contexts. It analyses the landscape of recent research, identifies key challenges, and introduces a structured framework to map LLM usage across six major domains: vulnerability detection, anomaly detection, cyber threat intelligence, blockchain security, penetration testing, and digital forensics. The review also examines the role of LLMs in cyber security education, highlighting how these models are being leveraged for personalized learning, hands-on simulation, and awareness-building. The key contribution of this paper lies in its holistic and integrative perspective, which distinguishes it from the existing literature. Prior reviews have tended to focus either on narrow technical implementations or on isolated use cases, often overlooking broader operational workflows and educational impact. In contrast, this paper systematically categorizes LLM applications across a wide spectrum of cyber security functions and uniquely integrates the educational dimension, which is an area that remains underexplored in current surveys. Furthermore, the paper identifies open issues and proposes future research directions that emphasize hybrid AI–human approaches, privacy-preserving architectures, and domain-specific evaluation frameworks. By bridging technical depth with educational insight, the paper advances the State of the Art and lays a foundation for more responsible, effective, and inclusive LLM adoption in the cyber security field. To support this transition, policymakers should promote ethical AI literacy and governance standards, while curriculum designers should embed LLM tools through scaffolded, skill-aligned learning frameworks.

Future work should focus on the co-development of privacy-preserving and explainable LLM systems, particularly for deployment in regulated environments and educational settings. Institutions are encouraged to implement hybrid AI–human frameworks that

tackle the automation capabilities of LLMs while simultaneously promoting critical thinking and informed decision-making. Within the educational domain, the development of standardized curriculum integration blueprints and ethically grounded usage policies is essential to ensure equitable access, transparency, and the preservation of essential skills. Additionally, the establishment of open, domain-specific benchmarks is crucial to enable reproducible research and robust performance validation, guiding the evolution of LLMs in alignment with technical excellence and broader societal responsibilities.

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